MAT415 Lecture Notes

Arky!! :3c

’26 Fall Semester

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Day 1: Gaussian Integers (Sep. 8, 2026)

The Quercus site for this class will be up tonight or early tomorrow. This class will have (approximately) weekly homework. There will be regular oral quizzes with the TA, about 5–10 minutes, where you will discuss one of the problems on the homework.

Consider the following example,

$$1 + 1 = 2, \quad 1 + 4 = 5, \quad 4 + 9 = 13, \quad 1 + 16 = 17.$$

Theorem 1.1 (Fermat)**.** Let $p > 2$; p can be expressed as a sum of two squares if and only if $p \equiv 1 \pmod 4$.

Proof. We start with the easy direction. Observe that the squares of the numbers $0, 1, 2, 3, \ldots$ are all equal to 0 or 1 modulo 4, i.e., $0^2 = 0$, $1^2 = 1$, $2^2 = 4 \equiv 0 \pmod 4$, and $3^2 = 9 \equiv 1 \pmod 4$, and so on (you can always express a number as either $2n$ or $2n + 1$, then squaring, to check that this pattern continues). It follows that if $p = n^2 + m^2$, then $p \equiv 1 \pmod 4$; if it were the sum of the squares of two odd numbers, then p would be even, which contradicts it being prime.

For the other direction, consider the ring of Gaussian integers,

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\} \subset \mathbb{C}.$$

Consider $p = n^2 + m^2$. Notice that the right hand side cannot be factored as *integers*, but it could be factored in $\mathbb{Z}[i]$ as $(n + im)(n - im)$; in this manner, we may instead ask the question of how to factor $p \in \mathbb{Z}[i]$. The continuation of this proof will use the fact that $\mathbb{Z}[i]$ is a UFD, but we will not prove that today. In particular, $\mathbb{Z}[i]$ is a Euclidean domain with $\alpha \mapsto |\alpha|^2$ is a Euclidean norm. □

Given $z = x + iy \in \mathbb{Z}[i]$, define the norm of z as $N(z) = z \cdot \overline{z} = x^2 + y^2 \in \mathbb{Z}_{\geq 0}$.[1] Observe that we have $N(z \cdot w) = N(z) \cdot N(w)$.

Claim 1.2. $z \in \mathbb{Z}[i]$ is a unit if and only if $N(z) = 1$, i.e., the units are $\{\pm 1, \pm i\}$.

Proof. Suppose $N(z) = 1$; by definition, we see $z \cdot \overline{z} = 1$, and so $\overline{z}$ constitutes the inverse of z. Conversely, suppose $z \in \mathbb{Z}[i]$ is a unit; if there exists another number w such that $z \cdot w = 1$, then, since norm is multiplicative, we may take the norm of both sides to get that $N(z) \cdot N(w) = 1$. Since $N(z) \in \mathbb{Z}_{\geq 0}$, it must be that $N(z) = 1$. □

Claim 1.3. If p is a prime such that $p \equiv 1 \pmod 4$, then p is not prime in $\mathbb{Z}[i]$.

We will prove the claim later; for now, observe that 1.1 and 1.3 imply each other; one is obvious, and for the direction that the claim implies the theorem, suppose p is not prime, so we may write it as the product of two non-units, $p = \alpha \cdot \beta$, where $\alpha, \beta \in \mathbb{Z}[i]$. We have that $p^2 = N(p) = N(\alpha) \cdot N(\beta)$; but $N(\alpha), N(\beta) \neq 1$, so it must be that $N(\alpha) = N(\beta) = p$. By writing $\alpha = a + bi$, we get that $N(\alpha) = a^2 + b^2 = p$, which is the desired expression as seen in 1.1.

We now prove 1.3 in two ways. For the first way, consider Wilson's theorem:

Theorem 1.4 (Wilson)**.** Let p be prime; then $(p - 1)! \equiv -1 \pmod p$.

Proof. The proof of this is simple; since $\mathbb{Z}/p$ is a field, every nonzero residue modulo p has an inverse, where $\alpha \equiv \alpha^{-1}$ if and only if $\alpha \equiv 1$ or -1, i.e., $\alpha^2 = 1$. It follows that the product of $1, \ldots, p - 1$ without 1 and $p - 1$ is 1, so we are done. □

[1] at this point, Rozenblyum made a comment about complex number norms versus algebra norms. idk. sure.

Corollary 1.5. If $p = 4n + 1$, then $((2n)!)^2 \equiv -1 \pmod{p}$.

Proof. By Wilson's theorem, we have that $-1 \equiv (p-1)!$. Let us write the factorial out;

$$-1 \equiv (p-1)! = (1 \cdot 2 \cdot \ldots \cdot 2n)\big((p-2n)(p-2n+1)\ldots(p-1)\big).$$

Since we're working modulo p, we may realize the right hand side expression as

$$\cdots \equiv (1 \cdot 2 \cdot \ldots \cdot 2n)\big((-2n)(-2n+1)\ldots(-1)\big) = (2n)!(-1)^{2n}(2n)! = \big((2n)!\big)^2. \qquad \square$$

The above corollary allows us to write[2]

$$p \mid \big((2n)!\big)^2 + 1,$$

which we may factor as $((2n)! + i)((2n)! - i)$; if p was prime, then it divides one of them, but we know that $p(a + bi) = (2n)! \pm i$ can't happen.

We now proceed to the second proof. Consider the cyclic group

$$(\mathbb{Z}/p)^\times = \mathbb{Z}/(p-1);$$

if $p = 4n + 1$, then $\mathbb{Z}/(p-1) \cong \mathbb{Z}/(4n)$, whence the existence of some element of order 4 in $(\mathbb{Z}/p)^\times$. Let such an element be called m; it follows that $m^4 \equiv 1 \pmod{p}$ and $m^2 \equiv -1 \pmod{p}$, but then we would have $p \mid m^2 + 1$, which is a contradiction. While this proof was fast, it didn't really tell us much about the arithmetic that went on in the first proof.

The morals as to the two above proofs are that

(i) It is important to understand arithmetic in more general number rings (such as prime factorization). As an example,

$$\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$$

is not a UFD (see the MAT347 notes for a reference on this).

(ii) Questions about number rings are closely related to questions in modular arithmetic.

[2] Rozenblyum proceeded to stand stunned in front of the blackboard for 3 minutes. When pointed out on how to continue the discussion, he said that this was proof your "IQ drops precipitously when in front of a classroom".

Day 2: Algebra Preliminaries; Algebraic Numbers and Integers (Sep. 10, 2026)

We set some conventions for this class. Unless specified otherwise, a ring will always refer to a commutative ring with unity.

Definition 2.1 (Preliminaries)**.** Let R be a ring.

(i) R is an *integral domain* if it admits no zero divisors, i.e., whenever $ab = 0$ for $a, b \in R$, then $a = 0$ or $b = 0$.

(ii) We will write $a \mid b$ if "a divides b" for $a, b \in R$ if there exists some $c \in R$ such that $b = ac$.

(iii) An integral domain R is a *principal ideal domain* (PID) if every ideal is principal, i.e., generated by a single element. Specifically, if $I \subset R$ is an ideal, then there exists $a \in R$ such that $I = \langle a \rangle$.

(iv) An element $u \in R$ is a *unit* if it has a multiplicative inverse. An element $p \in R$ is *irreducible* if it is not a unit, and whenever $p = ab$, either a or b is a unit. An element $p \in R$ is *prime* if, whenever $p \mid ab$, then $p \mid a$ or $p \mid b$.

(v) An integral domain R is a *unique factorization domain* (UFD) if every non-unit $x \in R$ is uniquely (up to permutation and multiplication by units) a product of irreducible elements.

Proposition 2.2. If R is a UFD, then irreducibles are primes, and primes are irreducible.

Note that primes are always irreducible. You can reference my MAT347 notes for a proof on this.

Proposition 2.3. Let R be a ring. Then $p \in R$ is prime if and only if $R/\langle p \rangle$ is an integral domain.

Proof. Pick $\overline{a}, \overline{b} \in R/\langle p \rangle$ such that $\overline{ab} = 0$ in $R/\langle p \rangle$. Then $ab \in \langle p \rangle$, so $p \mid ab$, where $p \mid a$ or $p \mid b$ as p is prime, so $\overline{a} = 0$ or $\overline{b} = 0$. The converse direction is obtained by reversing this proof. □

We now continue writing out the definitions.

Definition 2.4 (Preliminaries, cont.)**.** Let R be a ring.

(vi) An integral domain R is a *Euclidean domain* (ED) if there exists a map $v \colon R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ such that, for all $a, b \in R$ with $b \neq 0$, there exists $q, r \in R$ such that $a = bq + r$ with $r = 0$ or $v(r) < v(b)$. In addition, for all nonzero $a, b \in R$, v must satisfy $v(a) \leq v(ab)$.

Theorem 2.5. Let R be an integral domain. If R is a Euclidean domain, then it is a PID. If R is a PID, then it is a UFD.

Once again, the proof is in my MAT347 notes. Let us give some examples of this.

- $R = \mathbb{Z}$ is a Euclidean domain with $v(n) = |n|$.
- If F is a field, then $F[x]$ is a Euclidean domain with $v(f(x)) = \deg f$.
- Let n be an integer, and let $R = \mathbb{Z}/n$ be the integers mod n. R is an integral domain if and only if n is prime; in fact, if n is prime, then $\mathbb{Z}/n$ is a field. For a quick proof of this fact, let $x \in \mathbb{Z}$; then $\overline{x} \in \mathbb{Z}/n$ is a unit if and only if x is coprime to n. If p is rpime, then $(\mathbb{Z}/p)^\times$ is a group of order $p - 1$.

Theorem 2.6. Let $p \in \mathbb{N}$ be a prime number.

(i) (Fermat's little theorem) If $a \in \mathbb{Z}$ is coprime to p, then $a^{p-1} \equiv 1 \pmod{p}$.

(ii) $(\mathbb{Z}/p)^\times$ is a cyclic group.

Corollary 2.7. Let $p \in \mathbb{N}$ be a prime number. Then there exists $g \in \mathbb{Z}$, called the *primitive root modulo* p, such that (i) if a is coprime to p, then $a \equiv g^r \pmod{p}$ for some $r \geq 0$, and (ii) if $r \in \mathbb{Z}$ such that $g^r \equiv 1 \pmod{p}$, then $p - 1 \mid r$.

Definition 2.8. An *algebraic number* α is a complex number $\alpha \in \mathbb{C}$ for which there exists $a_0, \ldots, a_{n-1} \in \mathbb{Q}$ (with $n \geq 1$) such that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0,$$

i.e., $p(\alpha) = 0$ for some monic $p(x) \in \mathbb{Q}[x]$. An *algebraic integer* is a complex number $\alpha \in \mathbb{C}$ such that there exists $a_0, \ldots, a_{n-1} \in \mathbb{Z}$ (with $n \geq 1$) such that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0,$$

i.e., $p(\alpha) = 0$ for some monic $p(x) \in \mathbb{Z}[x]$. Algebraic numbers are denoted by $\mathbb{Q}^{\text{alg}}$ or $\overline{\mathbb{Q}}$, and algebraic integers are denoted by $\mathbb{Z}^{\text{alg}}$ or $\overline{\mathbb{Z}}$. The bar denotes algebraic closure.

In particular, algebraic numbers form a field, and algebraic integers form a ring. This is not obvious from the definition, but we will take it as true for now. Let us give some examples.

- Any $\alpha \in \mathbb{Q}$ is an algebraic number, since we may pick $p(x) = x - \alpha = 0$. Similarly, any $n \in \mathbb{Z}$ is an algebraic integer per the exact same construction.
- $\mathbb{Z}^{\text{alg}} \subset \mathbb{Q}^{\text{alg}}$.
- $\sqrt{-5} \in \mathbb{Z}^{\text{alg}}$, since $(\sqrt{-5})^2 + 5 = 0$.
- $\omega = \frac{1}{2}(-1 + \sqrt{-3})$ an algebraic integer, since $\omega^2 + \omega + 1 = 0$.
- For $n \geq 1$, we have that $\zeta = \exp(2\pi i/n) \in \mathbb{Z}^{\text{alg}}$, since $\zeta^n - 1 = 0$. ζ is the *primitive* n^{th} *root of unity*.
- Let $\alpha \in \mathbb{Q}^{\text{alg}}$ and $r \in \mathbb{Q}$. Then $r\alpha \in \mathbb{Q}^{\text{alg}}$. We will see later that, for all $\alpha \in \mathbb{Q}^{\text{alg}}$, there exists some $r \in \mathbb{Z}$ such that $r\alpha \in \mathbb{Z}^{\text{alg}}$.
- $\mathbb{Q}^{\text{alg}}$ is countable, since it is the union of the (finitely many) roots of polynomials in $\mathbb{Q}[x]$, of which are countable. $\mathbb{C}$ is not countable, however, so $\mathbb{Q}^{\text{alg}} \subsetneq \mathbb{C}$.
- $e, \pi \notin \mathbb{Q}^{\text{alg}}$.

Another way to characterize $\mathbb{Q}^{\text{alg}}$ and $\mathbb{Z}^{\text{alg}}$ is to consider subsets $V \subset \mathbb{C}$ such that V is a vector space (respectively, a finitely generated abelian group) over $\mathbb{Q}$. As an example, $V = \mathbb{Q}$ is trivially a vector space over $\mathbb{Q}$, and $V = \{r + s\sqrt{-5} \mid r, s \in \mathbb{Q}\}$ is one as well. In particular, it contains the finitely generated abelian group $\{r + s\sqrt{-5} \mid r, s \in \mathbb{Z}\}$.

Lemma 2.9. Let $\alpha \in \mathbb{C}$. Then

(i) $\alpha \in \mathbb{Q}^{\text{alg}}$ if and only if there exists a nonzero finite dimensional vector space $V \subset \mathbb{C}$ over $\mathbb{Q}$ such that $\alpha V \subset V$.

(ii) $\alpha \in \mathbb{Z}^{\text{alg}}$ if and only if there exists a finitely generated abelian group $M \subset \mathbb{C}$ such that $\alpha M \subset M$.

Proof. We start by proving (i). In the forwards direction, suppose $\alpha \in \mathbb{Q}^{\text{alg}}$; then $\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0$. Let

$$V = \{r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1} \mid r_0, \ldots, r_{n-1} \in \mathbb{Q}\},$$

which we readily see as finite-dimensional as a vector space over $\mathbb{Q}$. It follows that $\alpha V \subset V$, since

$$\begin{aligned}\alpha(r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1}) &= r_0\alpha + r_1\alpha^2 + \cdots + r_{n-1}\alpha^n \\ &= r_0\alpha + r_1\alpha^2 + \cdots + r_{n-1}(-a_{n-1}\alpha^{n-1} - \cdots - a_0) \in V.\end{aligned}$$

Conversely, suppose $V \subset \mathbb{C}$ is finite-dimensional with $\alpha V \subset V$. Let $v_1, \ldots, v_d$ be a basis of V over $\mathbb{Q}$. For each i, we see that

$$\alpha v_i = \sum_{j=1}^{d} r_{ij} v_j, \qquad r_{ij} \in \mathbb{Q};$$

i.e., $v \mapsto \alpha v$ is represented by a matrix $R := (r_{ij})$ in this basis. Specifically, R is a $d \times d$ matrix with entries in $\mathbb{Q}$, which we may think of as a complex matrix:

$$R\begin{pmatrix} v_1 \\ \vdots \\ v_d \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^{d} r_{ij} v_j \\ \vdots \\ \sum_{j=1}^{d} r_{dj} v_j \end{pmatrix} = \alpha \begin{pmatrix} v_1 \\ \vdots \\ v_d \end{pmatrix}.$$

It follows that $\alpha \in \mathbb{C}$ is an eigenvalue of R, whence α is a root of the characteristic polynomial $\det(R - Ix) \in \mathbb{Q}[x]$, so it has to be an algebraic number.

The proof for (ii), i.e., the algebraic integers case, follows analogously. Note that we need to know that the finitely generated abelian group is free; per the classification theorem (of finitely generated abelian groups), every such V can be written in terms of its free part and torsion; but there is no torsion, so we may reiterate our earlier argument as desired. □

Proposition 2.10. $\mathbb{Q}^{\text{alg}} \subset \mathbb{C}$ is a subfield, and $\mathbb{Z}^{\text{alg}} \subset \mathbb{C}$ is a subring.

Proof. Let $\alpha, \beta \in \mathbb{Q}^{\text{alg}}$. Then, per 2.9, there exist nonzero finite-dimensional subspaces $V, W \subset \mathbb{C}$ such that $\alpha V \subset V$ and $\beta W \subset W$. Consider

$$V \cdot W = \{v_1 w_1 + \ldots v_t w_t \mid v_j \in V, w_j \in W\};$$

observing that $v_1, \ldots, v_n$ span V and $w_1, \ldots, w_m$ span W, we simply have that $v_i w_j$ for $1 \le i \le n$ and $1 \le j \le m$ span $V \cdot W$. Thus, $V \cdot W$ is finite-dimensional and nonempty.

Now, for any $v \in V$ and $w \in W$, we have that $\alpha \cdot (vw) = (\alpha v) \cdot w \in V \cdot W$, and $\beta \cdot (vw) = v \cdot (\beta w) \in V \cdot W$. By linearity for $z \in V \cdot W$, we have $\alpha z \in V \cdot W$ and $\beta z \in V \cdot W$, whence $(\alpha + \beta)(V \cdot W) \subset V \cdot W$ and $(\alpha \cdot \beta)(V \cdot W) \subset V \cdot W$, so $\alpha + \beta$ and $\alpha\beta$ are both algebraic numbers per 2.9. An analogous argument holds for $\alpha, \beta \in \mathbb{Z}^{\text{alg}}$.

It remains to show that nonzero $\alpha \in \mathbb{Q}^{\text{alg}}$ have inverses, i.e., $\alpha^{-1} \in \mathbb{Q}^{\text{alg}}$. Indeed, the "multiplying by α" map, $\alpha \colon V \to V$, is injective, so it is isomorphic, and so there exists α^{-1} as the inverse map to α. In particular, it preserves V. Alternatively, we may write

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0 \implies \alpha \cdot \left(-\frac{1}{a_0}\right)(\alpha^{n-1} + a_{n-1}\alpha^{n-2} + \cdots + a_1) = 1,$$

which is a more explicit construction for α^{-1}. □

We now look at more examples.

- Every element of $\mathbb{Z}[\sqrt{-5}]$ is an algebraic integer. Indeed, every $\alpha \in \mathbb{Z}[\sqrt{-5}]$ is of the form $a + b\sqrt{-5}$ with $a, b \in \mathbb{Z}$, and there is clearly an easy monic quadratic $p(x) \in \mathbb{Z}[x]$ such that $p(\alpha) = 0$.
- $\alpha = \sqrt{2} + \sqrt{3} \in \mathbb{Z}^{\text{alg}}$, since it satisfies $\alpha^4 - 10\alpha^2 + 1 = 0$.

Proposition 2.11. Let $\alpha \in \mathbb{Q}^{\text{alg}}$. Then there exists a unique monic irreducible polynomial $f(x) \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$. Moreover, if $g(x) \in \mathbb{Q}[x]$ also satisfies $g(\alpha) = 0$, then $f \mid g$.

Proof. Consider $I = \{h(x) \in \mathbb{Q}[x] \mid h(\alpha) = 0\} \neq \varnothing$ (nonempty, as α is algebraic). I is clearly an ideal, and $\mathbb{Q}[x]$ is a PID, so we may write $I = \langle f \rangle$ for some f. Since f is uniquely determined up to units, we may divide any choice of f by its leading coefficient to let f be monic. It remains to check that f is irreducible. Let $f = gh$; then $0 = f(\alpha) = g(\alpha)h(\alpha)$. It must be that one of $g(\alpha)$ or $h(\alpha)$ is zero, with the other being in I. Without loss of generality, let $g(x) \in I$; but then $f \mid g$, so $\deg g \geq \deg f$, whence h is a unit. □

Day 3: Minimal Polynomials (Sep. 15, 2026)

Office hours will be held from 3–4pm on Wednesdays in BA6110. Last time, we defined the algebraic numbers, $\mathbb{Q}^{\text{alg}}$, and the algebraic integers, $\mathbb{Z}^{\text{alg}}$. We also saw that $\mathbb{Q}^{\text{alg}}$ is a field, and $\mathbb{Z}^{\text{alg}}$ is a ring.

Proposition 3.1. Let $\alpha \in \mathbb{Q}^{\text{alg}}$ be nonzero; there is a unique monic irreducible polynomial $f(x) \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$. Moreover, if $g(x) \in \mathbb{Q}[x]$ such that $g(\alpha) = 0$, then $f \mid g$.

Definition 3.2. Let $\alpha \in \mathbb{Q}^{\text{alg}}$. Then the unique monic irreducible $f(x) \in \mathbb{Q}[x]$ with $f(\alpha) = 0$ is called the *minimal polynomial of* α. The *degree of* α is the degree of $f(x)$ itself.

If $\alpha \in \mathbb{Q}^{\text{alg}}$, write $\mathbb{Q}(\alpha) = \{h(\alpha) \mid h \in \mathbb{Q}[x]\}$. If $\alpha \in \mathbb{Z}^{\text{alg}}$, write $\mathbb{Z}[\alpha] = \{h(\alpha) \mid h \in \mathbb{Z}[x]\}$. Clearly, $\mathbb{Q}(\alpha) \subset \mathbb{C}$, and we call it the subring generated by $\mathbb{Q}$ and α in $\mathbb{C}$. Similarly, $\mathbb{Z}(\alpha)$ is the smallest subring of $\mathbb{C}$ containing α.

As an example, take $\mathbb{Q}(\sqrt{-5}) = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Q}\}$, and $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$.

Proposition 3.3. Let $\alpha \in \mathbb{Q}^{\text{alg}}$.

(i) $\mathbb{Q}(\alpha) = \mathbb{Q}[x]/p(x)$, where p is the minimal polynomial of α.

(ii) As a vector space over $\mathbb{Q}$, $\mathbb{Q}(\alpha)$ has basis $1, \alpha, \ldots, \alpha^{n-1}$, where $n = \deg(\alpha)$.

(iii) $\mathbb{Q}(\alpha)$ is a field.

Proof. Consider the map $\text{ev}_\alpha \colon \mathbb{Q}[x] \to \mathbb{Q}(\alpha)$, given by $h \mapsto h(\alpha)$. ev_α is a surjective ring homomorphism by definition of $\mathbb{Q}(\alpha)$ (cf. earlier), whence

$$\mathbb{Q}(\alpha) \cong \frac{\mathbb{Q}[x]}{\ker(\text{ev}_\alpha)}.$$

In particular, we saw that $\ker(\text{ev}_\alpha) = \{h \in \mathbb{Q}[x] \mid h(x) = 0\} = \langle f(x) \rangle$, where $f(x) = a_0 + \cdots + a_{n-1}x^{n-1} + x^n$, so we are done with (i). (ii) immediately follows from earlier. For (iii), pick $\beta \in \mathbb{Q}(\alpha)$ nonzero, and let $\beta = h(\alpha)$ with $h \in \mathbb{Q}[x]$, for which $f \nmid h$, since $h(\alpha) \neq 0$. By the division algorithm (or Bezout's lemma), there must exist polynomials $A, B \in \mathbb{Q}[x]$ such that $Af + Bh = 1$. Evaluating at α, we see that $A(\alpha)f(\alpha) = 0$, so $B(\alpha)h(\alpha) = 1$, so we have found an inverse $\beta^{-1} = B(\alpha) \in \mathbb{Q}(\alpha)$ as desired. □

Example 3.4. We give some examples of irreducible polynomials.

(i) The minimal polynomial of $\sqrt{-5}$ is $x^2 + 5$.[3]

(ii) Let $\omega = \frac{1}{2}(-1 + \sqrt{-3}) = \exp(2\pi i/3)$. Clearly, $\omega^3 = 1$. Since $x^3 - 1$ is not irreducible, we see that we may factor and obtain the actual minimal polynomial for ω to be $x^2 + x + 1$.

Is $\frac{1}{2} \in \mathbb{Z}^{\text{alg}}$? Recall Gauss' lemma; suppose R is a UFD. A polynomial $f(x) \in \mathbb{R}[x]$ is *primitive* if the GCD of its coefficients is 1; Gauss' lemma states that if $f, g \in R[x]$ is primitive, then fg is primitive as well. As a quick proof, suppose not; then there exists $p \in R$ prime such that $p \mid fg$; since R/p is an integral domain, let $\overline{f}, \overline{g} \in (R/p)[x]$ be the corresponding reductions. We have that $\overline{f} \cdot \overline{g} = 0$, but then it must be that $\overline{f} = 0$ or $\overline{g} = 0$, which is a contradiction.

[3] so... any idea why this is irreducible? "the eisenstein criterion?" *grimaces* wayyyy overkill...

Lemma 3.5. Let R be a UFD, with F its field of fractions. Suppose $g, h \in F[x]$ are monic such that $f \coloneqq gh \in R[x]$. Then $g, h \in R[x]$.

Proof. Let c, d be the LCM of the denominators of the coefficients of g, h. Then cg and dh are primitive polynomials in $R[x]$. Gauss' lemma would then imply that $cg \cdot dh = (cd) \cdot gh$ is a primtiive polynomial; but $gh \in R[x]$, so $cd \mid (cd) \cdot gh$ in $R[x]$, i.e., cd must be a unit in R. It follows that c and d are both units in R, whence $g, h \in R[x]$. □

Corollary 3.6. If $\alpha \in \mathbb{Z}^{\text{alg}}$, then its minimal polynomial lies in $\mathbb{Z}[x]$.

Proof. Let $p(x) \in \mathbb{Q}[x]$ be the minimal polynomial of α. Since $\alpha \in \mathbb{Z}^{\text{alg}}$, there exists a monic $h(x) \in \mathbb{Z}[x]$ such that $h(\alpha) = 0$. $p(x) \mid h(x)$, i.e., $h(x) = p(x)q(x)$, for some q with rational coefficients. From here, the lemma tells us that $p, q \in \mathbb{Z}[x]$. □

Corollary 3.7. $\mathbb{Q} \cap \mathbb{Z}^{\text{alg}} = \mathbb{Z}$.

Proof. Let $\alpha \in \mathbb{Q} \cap \mathbb{Z}^{\text{alg}}$. The minimal polynomial of α is $x - \alpha$, so $\alpha \in \mathbb{Z}$. □

Definition 3.8. Algebraic numbers $\alpha, \beta \in \mathbb{Q}^{\text{alg}}$ are *conjugate* if they have the same minimal polynomial.

Lemma 3.9. We have some more properties of minimal polynomials.

(i) If $f \in \mathbb{Q}[x]$ is monic and irreducible of degree d, then f has d distinct roots in $\mathbb{C}$.

(ii) If α is an algebraic number of degree d, then there exist exactly d algebraic numbers conjugate to α.

Clearly, (i) implies (ii). We will prove this next lecture.

Day 4: Number Fields (Sep. 17, 2026)

Office hours have been moved to 2–3pm on Thursdays. Last time, given an algebraic number $\alpha \in \mathbb{Q}^{\text{alg}}$, we defined the conjugates of α to be the algebraic numbers with the same minimal polynomial, and we introduced the following lemma,

Lemma 4.1. (i) Let $f \in \mathbb{Q}[x]$ be monic and irreducible of degree d; then f has d distinct roots in $\mathbb{C}$. (ii) If α is an algebraic number of degree d, then there exist exactly d algebraic numbers conjugate to α.

Proof. It is clear that (i) implies (ii), so it suffices to prove (i). First, $\mathbb{C}$ is algebraically closed, so $f = (x - \alpha_1) \dots (x - \alpha_d)$ with each $\alpha_i \in \mathbb{C}$. We need to show that $\alpha_i \neq \alpha_j$ for all i, j; suppose $\alpha \in \mathbb{C}$ is a root of f with multiplicity greater than 1. Then f being monic and irreducible implies that f is the minimal polynomial of α. By taking the derivative of f, we see that $f'(\alpha) = 0$; but then $\deg f' = d - 1$, which is a contradiction. □

Let's do some examples.

(i) The conjugates of $\sqrt{5}$ are $\sqrt{5}$ and $\sqrt{-5}$, since $x^2 - 5 = 0$ is the minimal polynomial of $\sqrt{5}$. Recall that it is easy to check if a quadratic polynomial is irreducible (check the rational roots).

(ii) Let $f(x) = x^4 - 10x^2 + 1$, which we know (from previous classes) may be factored as

$$f(x) = (x - \sqrt{2} - \sqrt{3})(x - \sqrt{2} + \sqrt{3})(x + \sqrt{2} - \sqrt{3})(x + \sqrt{2} + \sqrt{3}).$$

We claim that f is irreducible; there are many ways to check this, see the MAT347 notes. In my opinion, the easiest would be to check the extensions of $\mathbb{Q}$.

Corollary 4.2 (Viète)**.** If $\alpha \in \mathbb{Q}^{\text{alg}}$ has minimal polynomial $f(x) = x^d + a_{d-1}x^{d-1} + \dots + a_0$ with conjugates $\alpha_1, \dots, \alpha_d$, then

$$\alpha_1 \dots \alpha_d = (-1)^d a_0, \quad \alpha_1 + \dots + \alpha_d = -a_{d-1}.$$

The proof is immediate; just write $f(x) = (x - \alpha_1) \dots (x - \alpha_d)$.

Definition 4.3. An *algebraic number field* is a subfield of F of $\mathbb{C}$ which is finite-dimensional over $\mathbb{Q}$.

Equivalently, an algebraic number field is a finite extension of $\mathbb{Q}$. (recall from MAT347 that any finite extension is generated by a single element, i.e., the primitive element theorem. cf. D&F §14.4.25). If F is a number field, then the *ring of integers* of F is denoted $\mathcal{O}_F = F \cap \mathbb{Z}^{\text{alg}}$.

Remark 4.4. We have the following properties of number fields.

(i) Let $V = F \subset \mathbb{C}$ be a number field; then $\alpha \cdot V \subset V$ for $\alpha \in F$, and so $\alpha \in \mathbb{Q}^{\text{alg}}$. Specifically, any element of an algebraic number field is an algebraic number, $F \subset \mathbb{Q}^{\text{alg}}$.

(ii) $\mathcal{O}_F$ is a subring of $\mathbb{C}$ containing $\mathbb{Z}$.

(iii) $\mathbb{Q} \cap \mathbb{Z}^{\text{alg}} = \mathbb{Z}$, i.e., $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$.

(iv) Let $\alpha \in \mathbb{Q}^{\text{alg}}$; we have that $\mathbb{Q}(\alpha) \subset \mathbb{C}$ is an algebraic number field. The primitive element theorem states that *every* number field is of this form.

Example 4.5. Let $d \in \mathbb{Z} \setminus \{0, 1\}$ be squarefree. Let $F = \mathbb{Q}(\sqrt{d}) = \{a + b\sqrt{d} \mid a, b \in \mathbb{Q}\}$. What is $\mathcal{O}_F$? Observe that

$$(a + b\sqrt{d})(a - b\sqrt{d}) = a^2 - db^2, \quad (a + b\sqrt{d}) + (a - b\sqrt{d}) = 2a,$$

so the minimal polynomial is $x^2 - 2ax + (a^2 - db^2)$. We want that $2a \in \mathbb{Z}$ and $a^2 - db^2 \in \mathbb{Z}$. If $d \equiv 2, 3 \pmod 4$, then a, b are actually integers (checked by bashing). If $d \equiv 1 \pmod 4$, then a, b are both integers, or half-integers. Thus,

$$\mathcal{O}_F = \begin{cases} \mathbb{Z} + \mathbb{Z}\sqrt{d}, & d = 2, 3 \pmod 4, \\ \mathbb{Z} + \mathbb{Z}\left(\frac{-1+\sqrt{d}}{2}\right), & d = 1 \pmod 4. \end{cases}$$

Note that, for $\alpha \in \mathbb{Z}^{\text{alg}}$ and $F = \mathbb{Q}(\alpha)$, $\mathcal{O}_F$ can be bigger than $\mathbb{Z}[\alpha]$.

Lemma 4.6. (i) If $\alpha \in \mathbb{Q}^{\text{alg}}$, there exists a nonzero integer $m \in \mathbb{N}$ such that $m\alpha \in \mathbb{Z}^{\text{alg}}$.

(ii) If F is a number field, then $F = \text{Frac}(\mathcal{O}_F)$.

Proof. We start by proving that (i) implies (ii). Since $\mathbb{Z} \subset \mathcal{O}_F$, we have that $\alpha = \frac{m\alpha}{m} \in \text{Frac}(\mathcal{O}_F)$. Note that he skips a lot of intuition here, but the point is that $\mathcal{O}_F$ contains the integers, so from (i), we may write α as a fraction in terms of elements of $\mathcal{O}_F$; since $\mathcal{O}_F = F \cap \mathbb{Z}^{\text{alg}}$, it is enough to have $\alpha \in \mathcal{O}_F$ to get (ii).

To prove (i), let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ (with $a_i \in \mathbb{Q}$) be the minimal polynomial of α. Then there exists m such that $ma_i \in \mathbb{Z}$ for all i; it follows that we may multiply $f(x)$ throughout by m^n to see that $m\alpha$ is a root of

$$x^n + ma_{n-1}x^{n-1} + m^2a_{n-2}x^{n-2} + \cdots + m^n a_0 \in \mathbb{Z}[x]. \qquad \square$$

We do more examples now.

(i) Consider the Gaussian integers $F = \mathbb{Q}(i)$, where $i = \sqrt{-1}$ and $-1 \equiv 3 \pmod 4$. Then $\mathcal{O}_F = \mathbb{Z}[i]$, for which $\mathcal{O}_F$ is a Euclidean domain with Euclidean norm $N(\alpha) = \alpha\overline{\alpha}$.

(ii) Let $\omega = \exp(2\pi i/3) \in \mathbb{C}$. We have that $F = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$. Since $-3 \equiv 1 \pmod 4$, we have that $\mathcal{O}_F = \mathbb{Z} + \mathbb{Z}\omega = \mathbb{Z}[\omega] \neq \mathbb{Z}[\sqrt{-3}]$. We claim that this is a Euclidean domain, with $N(\alpha) = \alpha\overline{\alpha}$ the Euclidean norm.

(iii) Let $F = \mathbb{Q}\sqrt{-5}$. $-5 \equiv 3 \pmod 4$, so $\mathcal{O}_F = \mathbb{Z}[\sqrt{-5}]$. Consider the Euclidean norm $N(a + b\sqrt{-5}) = a^2 + 5b^2 \in \mathbb{Z}_{\geq 0}$.

Lemma 4.7. The only units in $\mathbb{Z}[\sqrt{-5}]$ are -1 and 1. The elements $3, 7, 1 + 2\sqrt{-5}, 1 - 2\sqrt{-5}$ are all irreducible.

Proof. We claim that $\alpha \in \mathbb{Z}[\sqrt{-5}]$ is a unit if and only if $N(\alpha) = 1$. Suppose $\alpha\beta = 1$; then $N(\alpha\beta) = N(\alpha)N(\beta) = 1$. Since $N(\alpha) \in \mathbb{Z}_{\geq 0}$, if $N(\alpha) = 1$, then $\alpha\overline{\alpha} = 1$, meaning $\overline{\alpha} = \alpha^{-1}$. In this manner, it is enough to solve $a^2 + 5b^2 = 1$, from which we get $b = 0$ and $a = \pm 1$, whence the units are ± 1.

We now check that our claimed elements are irreducible. Suppose 3 is not irreducible. Let us write $3 = \alpha\beta$, where α, β are non-units. Per above, we see that this must mean that their norms are not 1; so

$$9 = N(3) = N(\alpha)N(\beta) \implies N(\alpha) = N(\beta) = 3;$$

since $\alpha = a + b\sqrt{-5}$ with $N(\alpha) = a^2 + 5b^2 \neq 3$, we get a contradiction, and we are done. The process is analogous to check that 7 is irreducible, i.e., we arrive at a contradiction when we find that $a^2 + 5b^2 = 7$ has no integer solutions.

To check that $1 \pm 2\sqrt{-5}$ is irreducible (we'll only check the $1 + 2\sqrt{-5}$ case because the other follows similarly), suppose $1 + 2\sqrt{-5} = \alpha\beta$, where α, β are non-units. Since

$$21 = N(1 + 2\sqrt{-5}) = N(\alpha)N(\beta),$$

we must have $N(\alpha)$ is 3 or 7, which is not possible as we saw earlier. Thus, we are done. □

The moral of this situation is that, to analyze Diophantine equations, we really just want to bash things out with factorization. In addition, norms are really useful for testing these properties as well!

We recall some important properties of field extensions (perhaps, from MAT347). Let L/F be a finite extension (i.e., $F \subset L$). We denote the degree of the extension as

$$\deg L/F = |L : F| \coloneqq \dim_F L.$$

As an example, $|\mathbb{Q}(\alpha) : \mathbb{Q}| = \deg \alpha$. The tower law states that, if L/M and M/F are finite extensions, and $\omega_1, \ldots, \omega_n$ is a basis for L/M, then

$$L = M\omega_1 \oplus \cdots \oplus M\omega_n.$$

We also have

$$|L : F| = |L : M|\,|M : F|\,.$$

We have the following key example. Let F be a number field, and take $\alpha \in F$; then

$$|F : \mathbb{Q}| = |F : \mathbb{Q}(\alpha)|\,|\mathbb{Q}(\alpha) : \mathbb{Q}|\,,$$

for which the latter is simply just $\deg \alpha$. In particular, $\deg \alpha \mid |F : \mathbb{Q}|$. What can we do with this? Suppose $\alpha = \sqrt{2} + \sqrt{3}$. $\mathbb{Q}(\alpha)$ contains $\sqrt{2}$ and $\sqrt{3}$. Since

$$\sqrt{2} = \frac{\alpha^3 - 9\alpha}{-4},$$

we see that $\mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\alpha)$. By writing

$$\deg \alpha = |F : \mathbb{Q}| = \left|F : \mathbb{Q}(\sqrt{2})\right|\left|\mathbb{Q}(\sqrt{2}) : \mathbb{Q}\right| = 4,$$

where $|\mathbb{Q}(\sqrt{2}) : \mathbb{Q}| = 2$, we see that $|F : \mathbb{Q}(\sqrt{2})| = 2$.

We now discuss norm and trace.

Definition 4.8. Let L/F be a finite extension. The *norm* of α, denoted $N_{L/F}(\alpha)$, is the determinant of the F-linear map $\phi_\alpha = \bullet\alpha \colon L \to L$. Similarly, the trace of α is $\mathrm{Tr}_{F/L}(\alpha) \coloneqq \mathrm{tr}(\phi_\alpha)$.

Let us look at some examples.

(i) $\mathbb{C}/\mathbb{R}$ has basis $\{1, i\}$. The matrix for multiplication by $x + iy$ is

$$\begin{pmatrix} x & -y \\ y & x \end{pmatrix},$$

(which you may perhaps recognize as the Cauchy–Riemann equations). Then $N_{\mathbb{C}/\mathbb{R}}(x + iy) = x^2 + y^2$, and $\mathrm{Tr}_{\mathbb{C}/\mathbb{R}}(x + iy) = 2x$.

(ii) $\mathbb{Q}(i)/\mathbb{Q}$ follows the exact same process. The norm and trace of $a + bi$ is $a^2 + b^2$, $2a$ respectively, since the basis and matrix are the same.

(iii) $\mathbb{Q}(\omega)/\mathbb{Q}$, where $\omega^2 + \omega + 1 = 0$ (and $\omega = \exp(2\pi i/3)$), has basis $1, \omega$. Multiplication by $a + b\omega$ in this basis is

$$\begin{pmatrix} a & -b \\ b & a-b \end{pmatrix},$$

so $N_{\mathbb{Q}(\omega)/\mathbb{Q}}(a + b\omega) = a^2 + b^2 - ab$, and $\mathrm{Tr}_{\mathbb{Q}(\omega)/\mathbb{Q}}(a + b\omega) = 2a - b$. The norm is the product of the conjugates, and conjuagte in this situation is the complex conjugate (really, this has to do with the fact that the degree is 2; not every degree 2 extension is like this, though).

Proposition 4.9. We give the main properties of the norm here. $N_{L/F} \colon L \to F$ has the following properties,

(i) $N_{L/F}(\alpha_1\alpha_2) = N_{L/F}(\alpha_1)N_{L/F}(\alpha_2)$;

(ii) $N_{L/F}(a) = a^{|L:F|}$ if $a \in F$;

(iii) $N_{L/F}(\alpha) = 0$ if and only if $\alpha = 0$;

(iv) If M is intermediate between L and F, then

$$N_{L/F} = N_{M/F} \circ N_{L/M}.$$

Proof. Properties (i) to (iii) are clear, so we will check (iv) only. We will start with a proof sketch in the special case that $\alpha \in M$; let $\omega_1, \dots, \omega_n$ be a basis for L/M. Then $L = M\omega_1 + \cdots + M\omega_n$, and α acts block-diagonally. Thus, $N_{L/F}(\alpha) = N_{M/F}(\alpha)^n = N_{M/F} \circ N_{L/M}(\alpha)$. Perhaps, this will be continued later? □

Proposition 4.10. $\mathrm{Tr}_{L/F}$ has the following properties,

(i) $\mathrm{Tr}_{L/F}(\alpha_1 + \alpha_2) = \mathrm{Tr}_{L/F}(\alpha_1) + \mathrm{Tr}_{L/F}(\alpha_2)$;

(ii) $\mathrm{Tr}_{L/F}(a\alpha) = a\,\mathrm{Tr}_{L/F}(\alpha)$ for $a \in F$ and $\alpha in L$;

(iii) $\mathrm{Tr}_{L/F}(a) = |L : F|\, a$ if $a \in F$;

(iv) If M is an intermediate field between L and F, then

$$\mathrm{Tr}_{L/F} = \mathrm{Tr}_{M/F} \circ \mathrm{Tr}_{L/M}\,.$$

Proof. The proof is the same. □

Remark 4.11. Let L/F be Galois. Then

$$N_{L/F}(\alpha) = \prod_{\sigma \in \mathrm{Gal}(L/F)} \sigma(\alpha), \quad \mathrm{Tr}_{L/F}(\alpha) = \sum_{\sigma \in \mathrm{Gal}(L/F)} \sigma(\alpha).$$

In general, sum and products embed L into F. I didn't catch what he said here; he'll probably restate this next class, so, we'll see then.

We will do this next time.

Day 5: Norm and Trace (Sep. 22, 2026)

Recall, from last time, that we defined norm and trace for field extensions. Specifically, let K/F be a finite field extension; then, for $\alpha \in L$, the left-multiplication map $\phi_\alpha \colon L \to L$ given by $\beta \mapsto \alpha\beta$ si a K-linear map of K-vector spaces, for which

$$N_{L/F}(\alpha) = \det(\phi_\alpha), \quad \mathrm{Tr}_{L/F}(\alpha) = \mathrm{tr}(\phi_\alpha).$$

Note that this definition than that of the book (cf. p.15), so we will relate our definition to theirs now.

Proposition 5.1. Let $\overline{F}$ be an algebraic closure of F. Then, for $\alpha \in L$, we have that

$$\mathrm{Tr}_{L/F}(\alpha) = \sum_{\substack{\sigma \colon L \to \overline{F} \\ F\text{-embeddings}}} \sigma(\alpha), \quad N_{L/F}(\alpha) = \prod_{\substack{\sigma \colon L \to \overline{F} \\ F\text{-embeddings}}} \sigma(\alpha).$$

Proof. Observe that $F \subset F(\alpha) \subset L$. Then $\mathrm{Tr}_{L/F}(\alpha) = |L : F(\alpha)| \, \mathrm{Tr}_{F(\alpha)/F}(\alpha)$, where $F(\alpha) = F[x]/p(x)$, with $p(x)$ being the minimal polynomial of α. $p(x)$ has $n = \deg p = |F(x) : F|$ distinct roots, which we will call $\alpha_1, \ldots, \alpha_n \in F$, so there are exactly n embeddings of $F(\alpha) \hookrightarrow \overline{F}$. □

In particular,

$$\mathrm{Tr}_{F(\alpha)/F}(\alpha) = \sum_{i=1}^{n} \alpha_i, \quad N_{F(\alpha)/F}(\alpha) = \prod_{i=1}^{n} \alpha_i.$$

In the basis of $F(\alpha)$, i.e., $\{1, \alpha, \ldots, \alpha^{n-1}\}$, the matrix of ϕ_α is given by

$$\begin{pmatrix} 0 & & & -a_0 \\ 1 & & & -a_1 \\ & \ddots & & \vdots \\ 0 & & 1 & -a_{n-1} \end{pmatrix},$$

i.e., the companion matrix for the minimal polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ of α.

In L/F, there are $|L : F|$ F-algebra embeddings $L \to \overline{F}$, where each one takes α to one of $\alpha_1, \ldots, \alpha_n$.

Claim 5.2. Each α_i appears exactly $\frac{1}{n} |L : F|$ times.

Proof. For every embedding $F(\alpha) \to \overline{F}$, there exists $\frac{1}{n} |L : F|$ $F(\alpha)$-algebra embeddings $L \to \overline{F}$. Specifically,

$$|L : F(x)| = \frac{|L : \overline{F}|}{|F(\alpha) : F|} = \frac{L : F}{n}.$$ □

In this way, we may write

$$\sum_{\substack{\sigma \colon L \to \overline{F} \\ F\text{-alg}}} \sigma(\alpha) = n \sum_{\substack{\sigma \colon L \to \overline{F} \\ F\text{-alg}}} \sigma(\alpha) = n \, \mathrm{Tr}_{F(\alpha)/F}(\alpha) = \mathrm{Tr}_{L/F}(\alpha).$$

The same applies to norm. As a corollary, if L/F is a finite Galois extension with Galois group G, then, for $\alpha \in L$,

$$\mathrm{Tr}_{L/F}(\alpha) = \sum_{\sigma \in G} \sigma(\alpha), \quad N_{L/F}(\alpha) = \prod_{\sigma \in G} \sigma(\alpha).$$

Proposition 5.3. Let F be a number field. If $\alpha \in \mathcal{O}_F$, then $\mathrm{Tr}_{F/\mathbb{Q}}(\alpha), N_{F/\mathbb{Q}}(\alpha) \in \mathbb{Z}$.

Proof. We saw, by Viète's formulas, that

$$\mathrm{Tr}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha) = -a_{n-1}, \quad N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha) = (-1)^n a_0,$$

for which a_0, a_{n-1} are coefficients of the minimal polynomial of α, whence they are integers. It follows that

$$\mathrm{Tr}_{F/\mathbb{Q}}(\alpha) = |F : \mathbb{Q}(\alpha)| \,\mathrm{Tr}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha), \quad N_{F/\mathbb{Q}}(\alpha) = |F : \mathbb{Q}(\alpha)| \, N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha)$$

yields the desired result, as $|F : \mathbb{Q}(\alpha)|$ is an integer as well. □

We now talk about another important invariant: the discriminant (cf. p.18).

Definition 5.4. Let L/F be a finite extension, and pick $\alpha_1, \dots, \alpha_n \in L$; then the associated discriminant is

$$\Delta(\alpha_1, \dots, \alpha_n) \coloneqq \det(\mathrm{Tr}_{L/F}(\alpha_i \alpha_j))_{ij} \in F.$$

As an example, let $d \in \mathbb{Z} \setminus \{0, 1\}$ not be a cube, and consider $F = \mathbb{Q}(d^{1/3})$; then the minimal polynomial is $x^3 - d = 0$, so the basis for $F/\mathbb{Q}$ is $\{1, d^{1/3}, d^{2/3}\}$. We have that $\Delta(1, d^{1/3}, d^{2/3})$ may be computed as the determinant

$$\det \begin{pmatrix} \mathrm{Tr}_{F/\mathbb{Q}} 1 & \mathrm{Tr}_{F/\mathbb{Q}} d^{1/3} & \mathrm{Tr}_{F/\mathbb{Q}} d^{2/3} \\ \mathrm{Tr}_{F/\mathbb{Q}} d^{1/3} & \mathrm{Tr}_{F/\mathbb{Q}} d^{2/3} & \mathrm{Tr}_{F/\mathbb{Q}} d \\ \mathrm{Tr}_{F/\mathbb{Q}} d^{2/3} & \mathrm{Tr}_{F/\mathbb{Q}} d & \mathrm{Tr}_{F/\mathbb{Q}} d^{4/3} \end{pmatrix} = \det \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 3d \\ 0 & 3d & 0 \end{pmatrix} = -27d^2.$$

Note that, if F is a number field, and $\alpha_1, \dots, \alpha_n \in \mathcal{O}_f$, then $\Delta(\alpha_1, \dots, \alpha_n) \in \mathbb{Z}$ (the determinant of an integer-valued matrix is an integer).

Proposition 5.5. The discriminant detects bases. Let L/F be a finite extension, and let $\alpha_1, \dots, \alpha_n \in L$.

(i) If $\Delta(\alpha_1, \dots, \alpha_n) \neq 0$, then $\alpha_1, \dots, \alpha_n$ are linearly independent over F.

(ii) If $\alpha_1, \dots, \alpha_n$ form a basis for L/F and $\mathrm{char}\, F = 0$, then $\Delta(\alpha_1, \dots, \alpha_n) \neq 0$.

Proof. For (i), suppose $\alpha_1, \dots, \alpha_n$ are linearly dependent. Then there exists $a_1, \dots, a_n \in F$ such that $a_1\alpha_1 + \cdots + a_n\alpha_n = 0$, so for any j,

$$0 = \mathrm{Tr}_{L/F}\left(\alpha_j \left(\sum_i a_i \alpha_i\right)\right) = \sum_{i=1}^n a_i \,\mathrm{Tr}_{L/F}(\alpha_i \alpha_j) \implies \left(\mathrm{Tr}_{L/F}(\alpha_i \alpha_j)\right)_{ij} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = 0.$$

Thus, $\det(\mathrm{Tr}_{L/F}(\alpha_i \alpha_j)) = 0$.

For (ii), suppose $\{\alpha_1, \dots, \alpha_n\}$ is a basis of L/F and $\Delta(\alpha_1, \dots, \alpha_n) = c$. Then the F-linear map $\phi \colon L \to L$ given by $(\mathrm{Tr}_{L/F}(\alpha_i \alpha_j))$ has nonzero kernel (by reading the matrix as a linear transformation), and we can run the same argument backwards to see that there is some choice of $a_1, \dots, a_n \in F$, not all zero, such that

$$\sum_{i=1}^n a_i \,\mathrm{Tr}_{L/F}(\alpha_i \alpha_j) = 0$$

for all j. Now, set $\alpha = a_1\alpha_1 + \cdots + a_n\alpha_n \in L$, and consider $\beta \in L$ given by $\beta = b_1\alpha_1 + \cdots + b_n\alpha_n$; we have that

$$\mathrm{Tr}_{L/F}(\beta\alpha) = \mathrm{Tr}_{L/F}\left(\sum_{j=1}^n b_j \alpha_j \alpha\right) = \sum_{j=1}^n b_j \,\mathrm{Tr}_{L/F}(\alpha_j \alpha) = \sum_{j=1}^n b_j \left(\sum_{i=1}^n a_i \,\mathrm{Tr}_{L/F}(\alpha_i \alpha_j)\right) = 0.$$

If $\alpha \neq 0$, we may take $\beta = \alpha^{-1}$ to get $\mathrm{Tr}_{L/F}(1) = 0$, which is a contradiction, and we are done. [4] □

[4] this is why you don't rush proofs man.

Day 6: Integral Basis (Sep. 24, 2026)

Rozenblyum isn't particularly happy with how he introduced the discriminant, so we will be doing some linear algebra exposition for the start of this lecture. Consider a vector space V over F. A *symmetric bilinear form* is a map

$$\langle -, - \rangle : V \times V \to F$$

such that $\langle v, w \rangle = \langle w, v \rangle$ and, for all $v \in V$, the map $V \to F$ given by $w \mapsto \langle v, w \rangle$ is a linear map. $\langle -, - \rangle$ is *nondegenerate* if, for all $v \in V$, there exists some $w \in V$ such that $\langle v, w \rangle \neq 0$. Our main example is the following: consider L/F a finite field extension; then

$$L \times L \to F, \quad (\alpha, \beta) \mapsto \operatorname{Tr}(\alpha\beta)$$

is a symmetric bilinear form. Moreover, if $\operatorname{char} F = 0$, it's nondegenerate, since $\operatorname{Tr}(\alpha\alpha^{-1}) = \operatorname{Tr}(1) = |L : F|$. Suppose V/F is a vector space with symmetric bilinear form $\langle -, - \rangle$. If $\{v_1, \dots, v_n\}$ is a basis of V, we can form a matrix

$$Q = \left(\left\langle v_i, v_j \right\rangle\right)_{i,j}.$$

Observe that if $v = \sum_i a_i v_i$ and $w = \sum_j b_j v_j$, then

$$\langle v, w \rangle = \left\langle \sum_i a_i v_i, \sum_j a_j v_j \right\rangle = \sum_{i,j} a_i b_j \left\langle v_i, v_j \right\rangle = (a_1, \dots, a_n)\, Q \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

We may also observe that $\det Q \neq 0$ if and only if $\langle -, - \rangle$ is nondegenerate. Now, suppose $\{w_1, \dots, w_n\}$ is another basis of V. We have a change of basis matrix $P = (P_{ij})$, where

$$w_j = \sum_{i=1}^{n} P_{ij} v_i,$$

and we have $Q' = (\langle w_i, w_j \rangle)$ (i.e., representing the same symmetric bilinear form, but in a different basis). How are Q and Q' related?

$$\left\langle w_i, w_j \right\rangle = \left\langle \sum_{k=1}^{n} P_{ki} v_k, \sum_{\ell=1}^{n} P_{\ell j} v_\ell \right\rangle = \sum_{k,\ell} P_{ki} P_{\ell j} \left\langle v_k, v_\ell \right\rangle = \sum_{k,\ell} P_{ki} P_{\ell j} Q_{k\ell} = (P^\top Q P)_{ij},$$

i.e., $Q' = P^\top Q P$.

Corollary 6.1. Let L/F be a finite field extension, and let $\{\alpha_1, \dots, \alpha_n\}$ be a basis of L/F. Let $\{\beta_1, \dots, \beta_n\}$ be a different basis of L/F. Let P be the transition matrix from α to β; then

$$\beta_j = \sum_{i=1}^{n} P_{ij} \alpha_i,$$

and $\Delta(\beta_1, \dots, \beta_n) = (\det P)^2 \Delta(\alpha_1, \dots, \alpha_n)$.

Proof. Let Q be the matrix corresponding to the form $\operatorname{Tr}(\alpha\beta)$ in the basis $\{\alpha_1, \dots, \alpha_n\}$, i.e., $Q_{ij} = \operatorname{Tr}(\alpha_i \alpha_j)$. Then $\Delta(\alpha_1, \dots, \alpha_n) = \det Q$. Now, let Q' be the corresponding matrix to $\{\beta_1, \dots, \beta_n\}$; since $Q' = P^\top Q P$, we have that

$$\Delta(\beta_1, \dots, \beta_n) = \det Q' = \det P^\top \det Q \det P = (\det P)^2 \det Q = (\det P)^2 \Delta(\alpha_1, \dots, \alpha_n). \qquad \square$$

Recall that, if $p(x)$ is a polynomial of degree n with roots $\alpha_1, \ldots, \alpha_n$, then

$$\Delta(p) = \left(\prod_{i<j} (\alpha_j - \alpha_i) \right)^2$$

is a symmetric polynomial in the α_i. In particular, $\Delta(p)$ is a polynomial in $a_0, \ldots, a_{n-1}$ (i.e., the coefficients of p), and is called the *discrimnant of* p.

We will detour a little here to talk about elementary symmetric polynomials. Suppose $f(x_1, \ldots, x_n) \in F[x_1, \ldots, x_n]$. We say that f is symmetric if $f(x_{\sigma(1)}, \ldots, x_{\sigma(n)}) = f(x_1, \ldots, x_n)$ for all $\sigma \in S_n$. Specifically, let us define $\sigma_i(x_1, \ldots, x_n)$ to be the sums of proudcts of x_i's; i.e.,

$$\sigma_1 = x_1 + \cdots + x_n, \quad \sigma_2 = \sum_{i<j} x_i x_j, \quad \ldots \quad \sigma_n = x_1 \ldots x_n.$$

These are called the "elementary symmetric polynomials", for which the fundamental theorem of symmetric polynomials states that any symmetric $f(x_1, \ldots, x_n)$ can be expressed as $g(\sigma_1, \ldots, \sigma_n)$ for a unique g. This is why we're able to write $\Delta(p)$ as a polynomial in $a_0, \ldots, a_{n-1}$.

Theorem 6.2. Consider a field extension $L = F(\alpha)/F$, where α has minimal polynomial $p(x)$ of degree n. Then

$$\Delta(1, \alpha, \ldots, \alpha^{n-1}) = \Delta(p(x)).$$

Proof. Write $p(x) = (x-\alpha_1)\ldots(x-\alpha_n)$. We see $\mathrm{Tr}(\alpha) = \sum_{i=1}^n \alpha_i$ and $\mathrm{Tr}(\alpha^k) = \sum_{i=1}^n \alpha_i^k$, so we may use this fact to try and write the discriminant in terms of these traces. Recall that the disriminant of $\{1, \alpha, \ldots, \alpha^{n-1}\}$ is given by

$$\Delta(1, \alpha, \ldots, \alpha^{n-1}) = \det\left(\mathrm{Tr}_{L/F}\left(\alpha^{(i+1)+(j-1)}\right) \right)_{1 \le i,j \le n} = \det\left(\sum_{k=1}^n \alpha_k^{i+j-2} \right).$$

Denote the entries as $M_{i,j}$ for convenience. We wish to decompose M as $M = V^\top V$, where

$$V = \begin{pmatrix} 1 & \alpha_1 & \alpha_1^2 & \ldots & \alpha_1^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \ldots & \alpha_n^{n-1} \end{pmatrix};$$

note that V is called the *Vandermonde matrix*. Indeed, since $V_{ij} = \alpha_i^{j-1}$, we see that

$$(V^\top V)_{ij} = \sum_{k=1}^n V_{ik}^\top V_{kj} = \sum_{k=1}^n V_{ki} V_{kj} = \sum_{k=1}^n \alpha_k^{i-1} \alpha_k^{j-1} = M_{ij}$$

as desired. In this manner, it follows that 6.1 yields

$$\Delta(1, \alpha, \ldots, \alpha^{n-1}) = (\det M)(\det V)^2,$$

where $\det V = \prod_{i<j}(\alpha_j - \alpha_i)$, so the right hand side is precisely the discriminant of the polynomial $p(x)$ as seen earlier. □

As an observation,

$$\Delta(1, \alpha, \dots, \alpha^{n-1}) = N_{L/F}(p'(\alpha)).$$

Proof. Let $p(x) = (x - \alpha_1) \dots (x - \alpha_n)$, and denote the latter $n - 1$ terms as $g(x)$. We have that $p'(x)$ can be written $(x - \alpha_1)g'(x) + g(x)$, where

$$p'(\alpha_1) = g(\alpha_1) = \prod_{i>1} (\alpha_1 - \alpha_i).$$

We may let g take on $(x - \alpha_1) \dots \widehat{(x - \alpha_i)} \dots (x - \alpha_n)$ to also see that

$$p'(\alpha_i) = \prod_{j \neq i} (\alpha_i - \alpha_j),$$

so

$$N_{L/F}(p'(\alpha)) = \prod_{i=1}^{n} p'(\alpha_i) = \prod_{i \neq j} (\alpha_i - \alpha_j) = (-1)^{\frac{n(n-1)}{2}} \prod_{i<j} (\alpha_i - \alpha_j)^2,$$

where we picked up a term of $(-1)^{\#\{i<j\}}$ by counting. □

Definition 6.3. Let F be a number field. Let $I \subset \mathcal{O}_F$ be a nonzero ideal. An *integral basis of I* is a basis of $F/\mathbb{Q}$, $\{\omega_1, \dots, \omega_n\} \in F$, such that

(i) each $\omega_i \in I$,

(ii) $I = \mathbb{Z}\omega_1 + \dots + \mathbb{Z}\omega_n$.

An integral basis of F is an integral basis of $I = \mathcal{O}_F$.

Lemma 6.4. Let F be a number field. If $I \subset \mathcal{O}_f$ is a nonzero ideal, then I contains a basis of F.

Proof. Let $\alpha_1, \dots, \alpha_n \in F$ be a basis; then there exists $m \in \mathbb{Z}$ such that $m\alpha_i \in \mathcal{O}_F$. Let $p \in I$ be a nonzero element; then the left-multiplication by β map, $\alpha \mapsto \beta\alpha$ from $F \to F$ is invertible. In particular, this means $\{\beta(m\alpha_i)\}_i$ is a basis of F. □

Proposition 6.5. Let F be a number field and $I \subset \mathcal{O}_F$ a nonzero ideal. Then I has an integral basis.

Proof. Recall that, if $\omega_1, \dots, \omega_n \in \mathcal{O}_F$, then $\Delta(\omega_1, \dots, \omega_n) \in \mathbb{Z}$. Choose a basis of F, $\{\omega_1, \dots, \omega_n\} \in I$, such that $|\Delta(\omega_1, \dots, \omega_n)|$ is minimal (it can't be zero by linear independence). We claim that $\{\omega_1, \dots, \omega_n\}$ constitutes an integral basis.

Suppose not. Then there exists some element $\alpha \in I$ such that α is a rational linear combination of the basis elements, i.e., $\sum_i a_i\omega_i$ for $a_i \in \mathbb{Q}$ (but not all $a_i \in \mathbb{Z}$). Without loss of generality, suppose a_1 is not an integer. This means $a_1 = \lfloor a_1 \rfloor + \theta$ for some rational $\theta \in (0, 1)$. Consider

$$\omega_1' \coloneqq \alpha - \lfloor a_1 \rfloor \omega_1 = \theta\omega_1 + a_2\omega_2 + \dots + a_n\omega_n,$$

which is clearly in I, as it is still a $\mathbb{Q}$-linear combination. Clearly, $\{\omega_1', \omega_2, \dots, \omega_n\}$ is a basis for F. We now check that this contradicts our supposition: indeed,

$$\Delta(\omega_1', \omega_2, \dots, \omega_n) = \det(P)^2 \Delta(\omega_1, \dots, \omega_n), \quad P = \left(\begin{array}{c|ccc} \theta & & & 0 \\ a_2 & 1 & & \\ \vdots & & \ddots & \\ a_n & & & 1 \end{array}\right),$$

for which $\det P = \theta < 1$, whence $\left|\Delta(\omega_1', \omega_2, \dots, \omega_n)\right| < |\Delta(\omega_1, \dots, \omega_n)|$, so we are done. □

Proposition 6.6. Suppose $\{\omega_1, \ldots, \omega_n\}$ and $\{\omega_1', \ldots, \omega_n'\}$ are integral bases of I. Then the absolute values of the discriminants of both bases are equal, i.e.,

$$|\Delta(\omega_1, \ldots, \omega_n)| = \left|\Delta(\omega_1', \ldots, \omega_n')\right|.$$

In particular, $\omega_1, \ldots, \omega_n \in I$ is an integral basis if and only if it is a basis and the absolute value of the discriminant is minimal.

Proof. Without loss of generality, $|\Delta(\omega_1, \ldots, \omega_n)|$ is minimal. We have that $\mathbb{Z}^n \to I$ is an isomorphism under the maps

$$(a_1, \ldots, a_n) \mapsto \sum_i a_i \omega_i, \quad (b_1, \ldots, b_n) \mapsto \sum_i b_i \omega_i',$$

for which the change of basis matrix P gives an isomorphism $\mathbb{Z}^n \to \mathbb{Z}^n$. Any such change of basis must have $\det P = \pm 1$, so it follows that

$$\Delta(\omega_1', \ldots, \omega_n') = (\det P)^2 \Delta(\omega_1, \ldots, \omega_n) = \Delta(\omega_1, \ldots, \omega_n). \qquad \square$$

"An easy way to do exercise 5 on the homework is to use the above property."

Definition 6.7. Let F be a number field, and let $\{\omega_1, \ldots, \omega_n\} \in \mathcal{O}_f$ be an integral basis. The *absolute discriminant* of F is

$$\Delta_F = \Delta(\omega_1, \ldots, \omega_n).$$

By the 6.6, this is well-defined. As an example, take d to be a squarefree integer, and consider $F = \mathbb{Q}(\sqrt{d})$. We have that

$$\Delta_F = \begin{cases} 4d & d \equiv 2, 3 \pmod 4, \\ d & d \equiv 1 \pmod 4, \end{cases}$$

since $\Delta(1, \sqrt{d}) = 4d$ for the first case, and d for the second case because $\{1, \sqrt{d}\}$ is not an integral basis for F (we may consider $\frac{1}{2}(1 + \sqrt{d})$), and so Δ_F for $d \equiv 1 \pmod 4$ must be a square apart from $4d$, whence the only candidate is d (cf. p.143?).

This number has meaning... something called "ramification".

Definition 6.8. Let F be a number field, and let $I \subset \mathcal{O}_F$ be a nonzero ideal. The *norm of* I, $N(I)$, is defined as the number of elements in the quotient $\mathcal{O}_F/I$.

In order for the definition to make sense, we need to know that $\#\{\mathcal{O}_F/I\}$ is finite.

Proposition 6.9. Let F be a number field and $I \subset \mathcal{O}_F$ a nonzero ideal. Then $\mathcal{O}_F/I$ is finite, i.e., a ring with finitely many elements.

Proof. Let A, B be finitely generated abelian groups such that $A \subset B$. Then $\operatorname{rank} B/A = \operatorname{rank} B - \operatorname{rank} A$.

Alternatively, let $m \in I \cap \mathbb{Z}$ be nonzero (e.g., $\alpha \in I$ nonzero with minimal polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_0$; then $a_0 \in I$). Let $\alpha \in \mathcal{O}_F/I$. We can write $\alpha = \sum_i c_i \overline{w_i}$, where $c_i \in \mathbb{Z}$, with $c_i = mq_i + r_i$. Then

$$\alpha = \sum_i r_i w_i \mod I.$$

Since $m \in I$, we see that any element of $\mathcal{O}_F/I$ can be represented as $\sum_i r_i w_i$, for $r_i \in \mathbb{Z}$ and $|r_i| < m$. There are finitely such elements, so we are done. $\square$

Day 7: Dedekind Domains (Sep. 29, 2026)

Recall that $\mathbb{Z}[\sqrt{-5}] = \mathcal{O}_{\mathbb{Q}(\sqrt{-5})}$ is not a PID, since $3, 7, 1+2\sqrt{-5}, 1-2\sqrt{-5}$ are all irreducible elements, and

$$21 = 3 \cdot 7 = (1+2\sqrt{-5})(1-2\sqrt{-5}),$$

with ± 1 being units. In number rings, ideals generalize elements, and considering ideals in number rings restores unique factorization. We may think of these as "ideal numbers".

Recall, that for a ring R, with $I, J \subset R$ ideals, we can consider $I + J = \{a + b \in R \mid a \in I, b \in J\}$ and $IJ = \langle ab \mid a \in I, b \in J\rangle$. If $I = \langle a_1, \dots, a_n\rangle$ and $J = \langle b_1, \dots, b_m\rangle$, then $IJ = \langle a_i b_j\rangle$ for $i = 1, \dots, n$ and $j = 1, \dots, m$. In $\mathbb{Z}[\sqrt{-5}]$,

$$\begin{aligned}
\left\langle 3, 1+2\sqrt{-5}\right\rangle\left\langle 3, 1-2\sqrt{-5}\right\rangle &= \langle 3\rangle, \\
\left\langle 7, 1+2\sqrt{-5}\right\rangle\left\langle 7, 1-2\sqrt{-5}\right\rangle &= \langle 7\rangle, \\
\left\langle 3, 1+2\sqrt{-5}\right\rangle\left\langle 7, 1+2\sqrt{-5}\right\rangle &= \left\langle 1+2\sqrt{-5}\right\rangle, \\
\left\langle 7, 1-2\sqrt{-5}\right\rangle\left\langle 3, 1-2\sqrt{-5}\right\rangle &= \left\langle 1-2\sqrt{-5}\right\rangle.
\end{aligned}$$

In the above, the ideals on the left hand side are prime ideals in R. From this, we may multiply the top two examples to get

$$\langle 21\rangle = \left\langle 3, 1+2\sqrt{-5}\right\rangle\left\langle 3, 1-2\sqrt{-5}\right\rangle\left\langle 7, 1+2\sqrt{-5}\right\rangle\left\langle 7, 1-2\sqrt{-5}\right\rangle.$$

We want some general context in which the above is true.

Definition 7.1. A *Dedekind domain* is an integral domain R such that

(i) every nonzero prime ideal of R is maximal,

(ii) R is Noetherian, and

(iii) R is *integrally closed*, i.e., $R \subset \mathrm{Frac}(R)$. This means that, supposing $\alpha \in K$ such that $a_0, \dots, a_{n-1} \in R$ satisfy

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0,$$

i,e, α is a root of a monic polynomial $p(x) \in R[x]$, then $\alpha \in R$.

We now discuss some key facts of Dedekind domains.

Theorem 7.2. Every number ring $\mathcal{O}_F$ is a Dedekind domain.

Theorem. If R is a Dedekind domain and $I \subset R$ is a nonzero (proper) ideal, then there is a unique set of prime ideals, up to reordering, $p_1, \dots, p_n \subset R$ such that

$$I = p_1 p_2 \dots p_n.$$

As an example, if R is a PID, then R is a Dedekind domain.

Proof of 7.2. We proved that, if $I \subset \mathcal{O}_F$ is a nonzero ideal, then $\mathcal{O}_F/I$ is finite. This followed from the existence of integral bases. We now claim that if R is a finite integral domain, then it is a field.

Subproof. Take a nonzero element $x \in R$. Since R is an integral domain, the right-multiplication map $a \mapsto ax$ from $R \to R$ is injective automatically, and surjective since R is finite. Thus, there exists $y \in R$ such that $ay = 1$. ∎

From here, if $I \subset \mathcal{O}_F$ is a nonzero prime ideal, then $\mathcal{O}_F/I$ is a finite integral domain, which is a field, which means I is maximal. I prime implies that $\mathcal{O}_F/I$ is a finite PID.

Suppose $I_1 \subset I_2 \subset \dots$ is an ascending chain of ideals. If $I_i = 0$ for all i, then the chain is stationary. Otherwise, without loss of generality, suppose I_1 is nonzero. Then

$$R/I_1 \twoheadrightarrow R/I_2 \twoheadrightarrow R/I_3 \twoheadrightarrow \dots,$$

which must eventually be stationary, since we start with some finite underlying set in R/I_1, and the surjections $\twoheadrightarrow$ either decrease or keep the number of elements the same; in either case, there can only be so many maps decreasing the number of elements, meaning that the ascending chain must become stationary at some point.

To conclude that $\mathcal{O}_F$ is a Dedekind domain, we need to check that it is integrally closed. We will leave it as a homework exercise. To do this, you need to show that, for $\alpha \in \mathcal{O}_F$ such that $p(\alpha) = 0$, then $p \in \mathcal{O}_F[x]$ is monic, and so α is an algebraic integer. □

To prove 8.7, we want to do the following first:

Theorem. Let $I \subset R$ be a nonzero ideal in a Dedekind domain. Then there exists a nonzero ideal J such that IJ is principal.

Lemma 7.3. Let R be a Dedekind domain and $I \subset R$ a nonzero ideal. Then I contains a product of nonzero prime ideals.

Proof. Suppose not. Let S be the set of ideals of R which don't contain a product of prime ideals. Since R is Noetherian, S contains a maximal element $M \in S$. Observe that M is not prime, since M does not contain a product of prime ideals (i.e., we would have $M \subset M$ otherwise). In this way, there exists $a, b \in M$ such that $a, b \notin M$, but $ab \in M$. Consider $I = M + (a)$ and $J = M + (b)$, for which $I, J \not\subset M$. We have that $IJ \subset M$, since every element of IJ is either in M or (ab); in particular, this means $I, J \neq R$. Since M is maximal with $I, J \not\subset S$, we see that I, J must contain products of prime ideals, i.e.,

$$I \supset p_1 \dots p_n, \quad J \supset q_1 \dots q_m,$$

and so $M \supset IJ \supset (p_1 \dots p_n)(q_1 \dots q_m)$. □

We will prove 8.7 and 8.2 (these were relabeled because they were proved later) next lecture.

Day 8: Properties of Dedekind Domains, Ideal Class Group (Oct. 1, 2026)

Last time, we defined Dedekind domains. For now, an ideal will denote a *nonzero* ideal (just so Rozenblyum doesn't have to write nonzero every single time). Last lecture, we proved that every ideal in a Dedekind domain contains a product of prime ideals.

Lemma 8.1. R is a Dedekind domain with field of fractions K and $A \subsetneq R$ a proper ideal, then there exists some element $\gamma \in K \setminus R$ such that $\gamma A \subset R$.

Proof. Let $a \in A$ be a nonzero element; then $(a) \subset A$. By the earlier lemma,

$$A \supset (a) \supset p_1 \dots p_k,$$

with $p_1, \dots, p_k$ prime ideals (and k being chosen to be minimal). Every ideal is contained in a maximal ideal, so $A \subset P$, where P is a prime ideal (since maximal ideals are prime); so $p_1 \dots p_k \subset P$ (when you multiply ideals, you get a smaller ideal, which is why there is no contradiction).

We claim that $P = p_i$ for some i. Otherwise, for all i, there exists $a_i \in P_i \setminus P$; but $a_1 \dots a_k \in P$, so one of the a_i must be in P itself; from here, we would get that $P_i \subset P$, but then $P_i = P$, since prime ideals are maximal in Dedekind domains.

Without loss of generality, let $P = p_1$ per the above. This means we have

$$p_1 \dots p_i \subset (a) \subset A \subset P = p_1,$$

where, by minimality of k, $p_2 \dots p_k \not\subset (a)$; in other words, there exists some element $b \in (p_2 \dots p_k) \setminus (a)$. Take $\gamma = b/a \in K \setminus R$; we want to show that, for all $x \in A$, that $\gamma x \in R$. Equivalently, $bx \in (a)$. This is true for all $x \in p_1$, since $b \in p_2 \dots p_k$, so $bx \in p_1 \dots p_k \subset (a)$, so we are done. □

Theorem 8.2. If R is a Dedekind domain and $I \subset R$ an ideal, then there exists an ideal J such that IJ is principal.

Proof. Let $\alpha \in I$ be nonzero; then $(\alpha) \subset I$. Define $J = \{x \in R \mid xI \subset (\alpha)\}$; then $J \neq \varnothing$ is an ideal with $\alpha \in J$. We want to show that $IJ = (\alpha)$. It suffices to check the opposite direction, since $IJ \subset (\alpha)$ by construction; in this manner, let

$$A = \frac{1}{\alpha} IJ \subset R.$$

We want to check that $A = R$. Supposing not, then A would be a proper ideal, and by 8.1, there would exist $\gamma \in A \setminus R$ such that $\gamma A \subset R$. We'll show that $\gamma \in R$ to get a contradiction. It is enough to check that γ satisfies a monic polynomial with coefficients in R, since Dedekind domains are integrally closed.

Since $\alpha \in I$, we have that $J \subset A = \frac{1}{\alpha} IJ$, for which $\gamma J \subset \gamma A \subset R$. We claim that $\gamma J \subset J$, which we would see by demonstrating $\gamma xy \subset (\alpha)$ for $x \in J$ and $y \in I$. Indeed,

$$\frac{1}{\alpha} \gamma xy = \gamma \left(\frac{1}{\alpha} xy \right) \in R,$$

for which the bracketed portion is an element of A. We know that $\gamma A \subset R$ from earlier, so $\alpha^{-1} \gamma xy \in R$, which is precisely what we wanted to check.

In this way, $\gamma : J \to J$ is a map of abelian groups. Let $\alpha_1, \ldots, \alpha_m$ be a generating set for the ideal J. We get

$$\gamma \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_m \end{pmatrix} = M \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_m \end{pmatrix}$$

for M an $m \times m$ matrix over R, i.e., $\gamma\alpha_i = \sum_{i=1}^{m} M_{ij}\alpha_j$. γ is an eigenvalue of M, so γ is a root of the characteristic polynomial of M, i.e., a monic polynomial with coefficients in R. □

Define an equivalence relation on ideals $I \sim J$ if there exist $\alpha, \beta \in R$ such that $\alpha I = \beta J$. If $I \sim J \sim K$, then $\alpha I = \beta J$ and $\gamma J = \delta K$, so $\gamma\alpha I = \beta\gamma J = \beta\delta K$. Observe that $\sim$ is compatible with products, i.e., $I \sim I'$ and $J \sim J'$ yields $IJ \sim I'J'$. The upshot being, we may write $C\ell_R$ to be the "ideal classes" to denote the quotient set. This has a binary operation $[I] \cdot [J] = [IJ]$.

Claim 8.3. $C\ell_R$ is a group.

Proof. For any nonzero $\alpha \in R$, we have that $[\alpha I] = [(\alpha) \cdot I)] = [I]$, so (α) is clearly an identity element in $C\ell_R$. Note that, for any nonzero $\alpha, \beta \in R$, $[(\alpha)] = [(\beta)]$, since $\beta(alpha) = \alpha(\beta)$. This means that the identity element of $C\ell_R$ is given by equivalence class of principal ideals. Specifically, this means we may use 8.2 to find inverses. □

We call $C\ell_R$ the *ideal class group* of R. Note that, if R is a PID, then $C\ell_R = \{e\}$. Conversely, if the class group is trivial, then R is a PID as well; specifically, it is enough to show that, if $I \sim (a)$, then I is principal. $I \sim (a)$ is equivalent to there existing nonzero $\alpha, \beta \in R$ sucht hat $\alpha I = \beta(\alpha)$; but then

$$R \supset I = \frac{1}{\alpha}\beta(\alpha) \implies \frac{\beta\alpha}{\alpha} = \beta \in R.$$

We can regard $C\ell_R$ as a measure of how non-principal ideals in R can be.

Theorem 8.4. If $R = \mathcal{O}_F$, where F is a number field, then $C\ell_R$ is finite.

This will be proved next week. I will change the label and indexing when it is done.

We start by checking two properties of Dedekind domains.

Proposition 8.5. If A, B, C are ideals of a Dedekind ring R and $AB = AC$, then $B = C$.

Proof. There exists some ideal J such that $JA = (\alpha)$, for $\alpha \in R$. This means $JAB = JAC$, which means $\alpha B = \alpha C$, which we may divide throughout by α to get $B = C$. □

Proposition 8.6. If $A, B \subset R$ are ideals of a Dedekind domain R, then $A \mid B$ (i.e., there exists C such that $B = AC$) if and only if $B \subset A$.

Proof. If $A \mid B$, then $B = CA \subset A$. Suppose $B \subset A$; then there exists an ideal $J \subset R$ such that $JA = (\alpha)$ is principal. Take $C = \alpha^{-1}JB$; then clearly, $CA = B$. Since $B \subset A$, we have $JB \subset JA = (\alpha)$, so $\alpha^{-1}JB \subset R$. $\alpha^{-1}JB$ is an ideal, so we are done. □

Theorem 8.7. Every ideal in a Dedekind domain R is uniquely representable as a product of prime ideals.

Proof. We first check existence, i.e., that every ideal can be expressed as a product of prime ideals. Suppose not; then, since R is Noetherian, there exists a largest proper ideal I that cannot be expressed as a product of prime ideals. Every proper ideal is contained in a maximal ideal, so, in particular, $I \subset M$, for which M being maximal means it is prime. This means that there must exist some J such that $I = JM$. Note that $I \subsetneq J$, since, if $I = J$, then $J = RJ = MJ = 1$, meaning $R = M$, which is a contradiction.

By definition of I, we would have that $J = p_1 \dots p_k$ for prime ideals $p_1, \dots, p_k$. This would mean $I = p_1 \dots p_k M$ as a product of prime ideals, so we have our desired contradiction of our earlier supposition. This means every ideal *can* be expressed as a product of prime ideals.

We now check for uniqueness. Suppose $p_1 \dots p_k = q_1 \dots q_\ell$. Since, for all the prime ideals $p_1, \dots, p_k$, we have $p_1 \dots p_k \subset p_1$, we have $q_1 \dots q_\ell \subset p_1$, so there exists i such that $q_i = p_1$. Without loss of generality, let $i = 1$ by rearranging. By cancellation, $p_2 \dots p_k = q_2 \dots q_\ell$, and we may just induct on the maximum of $\{k, \ell\}$ to finish off the proof. □

With this, we can "do arithmetic with ideals". Given $I, J \subset R$, we can define things like the greatest common divisor and least common multiple,

$$\gcd(I, J) \coloneqq I + J, \quad \operatorname{lcm}(I, J) \coloneqq I \cap J.$$

Theorem 8.8. Let $I \subset R$ be an ideal in a Dedekind domain. Let $\alpha \in I$ be nonzero; then there exists $\beta \in I$ such that $I = (\alpha, \beta)$.

Proof. It is enough to find β such that $I = \gcd((\alpha), (\beta))$. Let $I = p_1^{n_1} \dots p_k^{n_k}$, where $p_1, \dots, p_k$ are distinct primes. Then, let $Q_1, \dots, Q_r$ be other distinct prime ideals that divide (α). We want $\beta \in \bigcap_i (p_i^{n_i} - p_i^{n_i+1})$, since this would imply that $\beta \in \bigcap_j (R - Q_j)$. Per the chinese remainder theorem, fix $\beta_i \in p_i^{n_i} - p_i^{n_i+1}$, and let β satisfy

$$\beta = \beta_i \pmod{p_i^{n_i+1}};$$

then $\beta = 1 \pmod{Q_j}$. (Recall that, in order to apply the chinese remainder theorem, we would need to show that any pair of ideals is comaximal; but $p_1^{n_i+1} + p_j^{n_j+1} = R$, for which $\gcd(p_i^{n_i+1}, p_j^{n_j+1}) = R$). □

Day 9: Ramification (Oct. 6, 2026)

Theorem 9.1. A Dedekind domain R is a UFD if and only if it is a PID.

Proof. If R is a PID, then it is a UFD. We will check the forwards direction. Suppose R is a UFD and $I \subset R$ is a nonzero ideal. We know that there exists $J \subset R$ such that $IJ = (a)$. Indeed, take the prime factorization $a = p_1 \dots p_r$ (and so $(a) = (p_1) \dots (p_r)$); since p_i being prime implies that the (p_i) are prime ideals, we see that

$$IJ = (p_1) \dots (p_r).$$

By uniqueness of prime factors of ideals, we see that $I = (p_{i_1}) \dots (p_{i_k}) = (p_{i_1} \dots p_{i_k})$, where $\{i_1, \dots, i_k\} \subset \{1, \dots, r\}$, so I is principal. □

We now discuss the splitting of primes. First, we already saw that primes in $\mathbb{Z}$ might factor in other number rings, i.e., in $R = \mathbb{Z}[i]$, $5 = (2 - i)(2 + i)$, but $3 \in \mathbb{Z}[i]$ is prime. There are three kinds of primes in the Gaussian integers. For example, 3 is prime in $\mathbb{Z}[i]$, 5 factors into distinct primes, and 2 factors as the *square* of a prime $(1 + i)^2$.

$R = \mathbb{Z}[\sqrt{-5}]$ is not a PID. $2, 3$ are irreducible in R, but they're not prime. Indeed,

$$(2) = (2, 1 + \sqrt{-5})(2, 1 + \sqrt{-5}), \quad (3) = (3, 1 + \sqrt{-5})(3, 1 - \sqrt{-5}).$$

We now discuss the more general setup. Let $K \subset L$ be number fields, and let $R \coloneqq \mathcal{O}_K$ and $S \coloneqq \mathcal{O}_L$ (so $R \subset S$).

Theorem 9.2. Let $P \subset R$, $Q \subset S$ be prime ideals; then the following are equivalent.

(i) Q divides into $PS = \{\alpha_1\beta_1 + \dots + \alpha_r\beta_k \mid \alpha_i \in P, \beta_j \in S\}$;

(ii) $PS \subset Q$;

(iii) $P \subset Q$;

(iv) $Q \cap R = P$;

(v) $Q \cap K = P$.

Proof. (i) is equivalent to (ii) because divisibility implies containment. (ii) is equiavlent to (iii) since PS is the smallest ideal of S containing P. (iv) is equivalent to (v) because $Q \cap K \subset R$. Clearly, (iv) implies (iii), so the only thing that remains to show is (iii) implies (iv). $P \subset Q$ implies $P \subset Q \cap R$; we may observe that $Q \cap R \subset R$ is a proper ideal, i.e., $1 \notin Q \cap R$, since 1 is not in Q. Since P is a nonzero prime, it is maximal, so $P = Q \cap R$. □

Definition 9.3. When conditions of the above theorem are satisfied, we say "Q lies over P", or "P lies under Q".

Theorem 9.4. In the context above,

(i) Every prime $Q \subset S$ lies over a unique prime $P \subset R$.

(ii) Every prime $P \subset R$ lies under at least one prime Q of S.

Proof. For (i), we need to show that $Q \cap R$ is a prime ideal. $ab \in Q \cap P$ implies that a or b is in $Q \cap R$. Since $1 \notin Q \cap R$, it follows that $Q \cap R$ is a proper ideal.

For (ii), take Q to be any prime ideal in the factorization os PS. We need to show that $PS \neq S$, i.e., $1 \notin PS$. Recall that there exists $\gamma \in K \setminus R$ such that $\gamma P \subset R$. This means $\gamma PS \subset S$; if $1 \in PS$, then $\gamma \in S$, meaning γ is an algebraic integer (by definition of S being a number field). γ is an algebraic integer contained in K, meaning it must be contained in $R = \mathcal{O}_K$, which is a contradiction. □

We now have that, if $p \subset R$ is prime, then $pS = q_1^{e_1} \dots q_r^{e_r}$, where the q_i's are distinct prime ideals over p, and the e_i are called *ramification indices*. p is said to be *ramified* if there are exponents greater than 1, i.e., some prime appears with multiplicity higher than 1.

If q lies over p, then the map $R \to S$ descends to a map

$$R/p \to S/q$$

in the quotient. Both R/p and S/q are fields, since p, q are maximal, meaning we S/q is a field extension of R/p. We denote the *inertial degree* of q over p as $|S/q : R/p|$, usually written $f(q/p)$.

As an example, consider $R = \mathbb{Z}$ and $S = \mathbb{Z}[i]$. 2 lies under $(1-i)$, and $S/2S$ has order 4, so $|S/(1-i)|$ is a proper divisor of 4, i.e., $|S/(1-i)| = 2$. On the other hand, 3 is a prime in $\mathbb{Z}[i]$. $3S \subset S$ is a prime ideal, and $|S/3S| = 9 = 3^2$, so $f(3S/3) = 2$.

With these examples, we see that

$$\sum_{i=1}^{r} e_i f_i = |L : K|,$$

where $f_i = f(q_i/p)$. As an example, for $\mathbb{Z} \subset \mathbb{Z}[i]$, $(2) = (1-i)^2$, so $e = 2$ and $f = 1$. Here, $|\mathbb{Q}(i) : \mathbb{Q}| = 2$. We also have $(3) = (3)$, so $e = 1$ and $f = 2$.

Day 10: Norms of Ideals, Decomposition of Extension of Number Fields (Oct. 8, 2026)

Let L/K be number fields, and let $p \subset \mathcal{O}_K$ be a prime ideal. Then $p\mathcal{O}_L$ has a decomposition as $q_1^{e_1} \dots q_r^{e_r}$, where the q_i are the primes lying over p, and e_i the ramification indices. Denote

$$f_i = |\mathcal{O}_L/q_i : \mathcal{O}_k/p|$$

to be the indices of inertia.

Theorem 10.1. In the context above, $\sum_{i=1}^{r} e_i f_i = n = |L : K|$.

To prove this, we will use the norm. Recall that, for $I \subset \mathcal{O}_K$ a nonzero ideal, we defined $N(I) = \#\mathcal{O}_K/I$ as the *norm of* I.

Theorem 10.2. For ideals $I, J \subset \mathcal{O}_K =: R$, $N(I \cdot J) = N(I)N(J)$.

Proof. Suppose I and J are relatively prime. Then

$$I + J = \gcd(I, J) = R,$$

so $I \cap J = \operatorname{lcm}(I, J) = IJ$. By the Chinese remainder theorem,

$$\frac{R}{I \cdot J} \cong \frac{R}{I} \times \frac{R}{J}.$$

Thus, $N(I \cdot J) = N(I)N(J)$ in this case.

It remains to check that if $p \subset \mathcal{O}_K$ is a prime ideal, then $N(p^k) = N(p)^k$. We will proceed by induction on k. The claim is clearly vacuously true for $k = 0$ or 1, so we will directly go to the inductive step. Let $S = R/p^k$; we have that $I = p^{k-1}/p^k$ is an ideal in S, so $S/I \cong R/p^{k-1}$. By counting elements, we get

$$\frac{\left|R/p^k\right|}{\left|p^{k-1}/p^k\right|} = \left|R/p^{k-1}\right| \implies \frac{N(p^k)}{\left|p^{k-1}/p^k\right|} = N(p^{k-1}),$$

i.e., $N(p^k) = N(p^{k-1})\left|p^{k-1}/p^k\right|$. By induction, we need to show that

$$\left|\frac{p^{k-1}}{p^k}\right| = N(p) = \left|\frac{R}{p}\right|.$$

We claim that there exists an isomorphism $p^{k-1}/p^k \cong R/p$ as abelian groups. To see this, pick $\beta \in p^{k-1} \setminus p^k$, and define $\varphi\colon R \to p^{k-1}/p^k$ by $\alpha \mapsto [\beta\alpha]$. We want to check that φ is surjective, and the kernel is exactly p. Notice that $\operatorname{im}\varphi \ni (\beta) + p^\alpha = \{\beta x + \alpha\}$ for $\alpha \in p^k$, where $(\beta) + p^k = p^{k-1}$, so $p^k \subsetneq (\beta) + p^k \subset p^{k-1}$. It follows that $(\beta) + p^k = p^{k-1}$ by uniqueness of prime factorization, so φ is surjective.

We now compute the kernel of φ. First, observe that $I = \{\alpha \mid \beta\alpha \in p^k\}$ is an ideal, where $p \subset I \subsetneq R$; but p is a maximal ideal, so $p = I$. This means $\ker\varphi = p$. □

Theorem 10.3. Let $\mathbb{Q} \subset L$ and $n \coloneqq |L : Q|$. Let $p \in \mathbb{Z}$ be prime. Given the decomposition $p\mathcal{O}_L = q_1^{e_1} \dots q_r^{e_r}$, we have

$$\sum_{i=1}^{r} e_i f_i = n, \quad f_i = |\mathcal{O}_L/q_i : \mathbb{Z}/p|\,.$$

Proof. Here, $N(p\mathcal{O}_L) = N(q_1)^{e_1} \dots N(q_r)^{e_r}$. Since $N(p\mathcal{O}_L) = p^n$ and $N(q_i) = p^{f_i}$, we may pick an integral basis $\alpha_1, \dots, \alpha_n$ such that

$$\frac{\mathcal{O}_L}{p\mathcal{O}_L} \cong \frac{\mathbb{Z}^n}{p\mathbb{Z}^n} = \left(\frac{\mathbb{Z}}{p}\right)^n.$$

Thus, $p^n = (p^{f_1})^{e_1} \dots (p^{f_r})^{e_r}$ yields the desired result by comparing powers. □

Theorem 10.4. Let L/K be a degree n extension of number fields. Let $p \subset \mathcal{O}_K$ be a prime ideal; then $N(p\mathcal{O}_L) = N(p)^n$.

Proof. Take $I = p_1^{e_1} \dots p_k^{e_k}$. By our previous theorem, it's enough to show this for $I = p$, i.e., a prime ideal p. Observe that $\mathcal{O}_L/p\mathcal{O}_L$ is a vector space over $\mathcal{O}_K/p\mathcal{O}_K$. We'll show that

$$\dim_{\mathcal{O}_K/p\mathcal{O}_K} \mathcal{O}_L/p\mathcal{O}_L = n.$$

Let's start by checking that the dimension is at most n, i.e., any collection of $n+1$ elements is linearly dependent. Let $\alpha_1, \dots, \alpha_{n+1} \in \mathcal{O}_L$; we want to show that $\overline{\alpha}_1, \dots, \overline{\alpha}_{n+1} \in \mathcal{O}_L/p\mathcal{O}_L$ are linearly dependent. First, $\dim_K L = n$, so the α_i's are linearly dependent over K. Thus, there exists $a_i \in K$ such that

$$\sum_{i=1}^{n+1} a_i \alpha_i = 0.$$

We can make $a_i \in \mathcal{O}_K$ by multiplying by large enough $N \in Z$. Recall, that, for all $x \in K$, there exists $n \in \mathbb{Z}$ such that $nx \in \mathcal{O}_K$. From here, we see that there exists $b_k \in \mathcal{O}_K$, not all in p, such that

$$\sum_{i=1}^{n+1} b_i \alpha_i = 0. \qquad \triangle$$

Before we do 10.1, we will prove a lemma first.

Lemma 10.5. Let $A, B \subset R$ be nonzero ideals in a Dedekind domain with $B \subset A \neq R$. Then there exists $\gamma \in \mathrm{Frac}(R)$ such that $\gamma B \subset R$, but $\gamma B \not\subset A$.

Proof. There exists $C \subset R$ such that $B \cdot C = (\alpha)$, i.e., principal. Then $B \cdot C \not\subset \alpha \cdot A$, so there exists $\beta \in C$ such that $\beta B \not\subset \alpha A$. Set $\gamma = \beta/\alpha$. We get that $\gamma B = \alpha^{-1}(\beta \cdot B) \subset \alpha^{-1}(C \cdot B) = \alpha^{-1}(\alpha) = R$. Notice that

$$(\gamma \cdot B) \subset A \iff \alpha\gamma \cdot B \subset \alpha A,$$

so $\alpha\gamma B = \beta B \not\subset \alpha A$. □

We now apply this lemma to finish off our earlier proof.

Proof of 10.4. Suppose $a_i \in P$ for all i. Applying our lemma to $A = P$ and $B = (a_1, \dots, a_{n+1}) \subset \mathcal{O}_K$, we get $\gamma \in K$ such that $\gamma B \subset \mathcal{O}_K$, i.e., $\gamma a_i \in \mathcal{O}_K$, but $\gamma B = (\gamma a_i) \not\subset A$. By taking $b_i = \gamma a_i \in \mathcal{O}_K$ (from earlier), we get

$$\sum_{i=1}^{n+1} b_i \alpha_i = \gamma \sum_{i=1}^{n+1} a_i \alpha_i = 0,$$

and we may reduce modulo P to get a linear dependence.

We want to check that the dimension (in the theorem statement) is n by checking the other direction of the inequality. Consider $P \subset \mathcal{O}_K$ a prime ideal, and let $P \cap \mathbb{Z} = \ell\mathbb{Z}$ for some prime $\ell \in \mathbb{Z}$. Consider all primes P_i in $\mathcal{O}_K$ lying over ℓ, i.e., $\ell\mathcal{O}_K \cong p_1^{e_1} \dots p_m^{e_m}$. We know

$$\dim_{\mathcal{O}_K/p_i} \mathcal{O}_L/p_i\mathcal{O}_L = n_i \leq n.$$

We'll show that $n_i = n$ for all i. We have that

$$\sum_i e_i f_i = d = |K : \mathbb{Q}|$$

for $f_i = f(P_i/\ell) = |\mathcal{O}_K/p_i/\mathbb{Z}/\ell|$; notice that $\ell\mathcal{O}_K$ has prime decomposition $p_1^{e_1} \dots p_k^{e_k}$; then $\ell\mathcal{O}_L \cong \prod(p_i\mathcal{O}_L)^{e_i}$, and

$$N(\ell\mathcal{O}_L) = N(p_1\mathcal{O}_L)^{e_1} \dots N(p_m\mathcal{O}_L)^{e_m} \leq (N(p_1)^{n_1})^{e_1} \dots (N(p_m)^{n_m})^{e_m}$$

by the previous theorem. Since $N(\ell\mathcal{O}_L) = \ell^{\deg|L:\mathbb{Q}|} = \ell^{dn}$ and $f_i = |\mathcal{O}_K/p_i : \mathbb{Z}/\ell|$, we see $N(p_i) = \ell^{f_i}$. Moreover, per the earlier theorem, we know $n_i \leq n$, so the inequality $N(p_m\mathcal{O}_L)^{e_m} \leq (N(p_m)^{n_m})^{e_m}$ becomes

$$\ell^{dn} = \ell^{\sum e_i f_i n_i} \implies \sum_i e_i f_i n_i = dn;$$

but $\sum e_i f_i = d$ and $n_i \leq n$. These three things can only be true if $n_i = n$, so we are done. □

We can finally return to prove the original statement presented at the start of class.

Proof of 10.1. Apply the norm N to $p\mathcal{O}_L = q_1^{e_1} \dots q_r^{e_r}$. Then

$$N(p\mathcal{O}_L) = N(q_1)^{e_1} \dots N(q_r)^{e_r}.$$

Since $f_i = |\mathcal{O}_L/q_i : \mathcal{O}_K/p|$, we get $N(p\mathcal{O}_L) = N(p)^n$ by the previous theorem. Also, $N(q_i) = N(p)^{f_i}$, so

$$N(p)^n = N(p)^{\sum_{i=1}^r e_i f_i} \implies n = \sum_{i=1}^r e_i f_i.$$ □

Day 11: Galois Extensions for Number Fields (Oct. 13, 2026)

Last time, we addressed how primes split in number rings, and we talked about norms of ideals.

Theorem 11.1. Let K be a number field. If $\alpha \in \mathcal{O}_K$ is nonzero, then $N(\langle\alpha\rangle) = N_K(\alpha)$.

As an abelian group, $\mathcal{O}_K/\langle\alpha\rangle = \mathcal{O}_K/\mathrm{im}(\varphi_\alpha)$, where $\varphi_\alpha \colon \mathcal{O}_K \to \mathcal{O}_\alpha$ is the group homomorphism given by left multiplication, $\varphi_\alpha(\beta) = \alpha\beta$.

Lemma 11.2. Let $\varphi \colon \mathbb{Z}^n \to \mathbb{Z}^n$ be a map of free abelian groups with $\det(\varphi) \neq 0$. Then

$$\left|\frac{\mathbb{Z}^k}{\mathrm{im}\,\varphi}\right| = |\det\varphi|\,.$$

Proof of 11.2. Let $P \in M_n(\mathbb{Z})$ be the matrix representing φ. By assumption, $\det P \neq 0$. We can now do row and column operations, i.e., Gaussian elimination, to transform P into a diagonal matrix. In this way, there exists elementary matrices $E_1, E_2 \in \mathrm{GL}_n(\mathbb{Z})$ and diagonal matrix $D = \mathrm{diag}(d_1, \ldots, d_n)$ such that

$$P = E_1 D E_2.$$

Then $\mathbb{Z}^n/P\mathbb{Z}^n = \mathbb{Z}^n/(E_1DE_2\mathbb{Z}^n) = E_1\mathbb{Z}^n/(DE_1\mathbb{Z}^n) = \mathbb{Z}^n/D\mathbb{Z}^n$. The number of elements in $\mathbb{Z}^n/D\mathbb{Z}^n$ is given by multiplying the number of elements in each $\mathbb{Z}/d_i\mathbb{Z}$, which is the product of all d_i, i.e., $|\det(\varphi)|$. □

Proof of 11.1. Let $\alpha_1, \ldots, \alpha_n$ be an integral basis of $\mathcal{O}_K$. In this basis, φ_α is given by an integer $n \times n$ matrix. The theorem then follows from the lemma. □

Now, consider L/K to be an extension of number fields, and take $p \subset \mathcal{O}_K$ a prime ideal. We saw that $p\mathcal{O}_L = q_1^{e_1} \ldots q_m^{e_m}$.

Theorem 11.3. Suppose L/K is a Galois (i.e., normal) extension. Then if q_1, q_2 are primes over $p \subset \mathcal{O}_K$, then there exists $\sigma \in \mathrm{Gal}(L/K)$ such that $\sigma(q_1) = q_2$.

Proof. Suppose not, i.e., $\sigma(q_1) \neq q_2$ for all σ. Then there exists $\alpha \in \mathcal{O}_L$ such that

$$\alpha \equiv 0 \pmod{q_2}, \quad \alpha \equiv 1 \pmod{\sigma(q_1)}$$

for all $\sigma \in \mathrm{Gal}(L/K)$. Note that the "modulo q_2" statement is equivalent to writing $\alpha \in p_2$. By the Chinese remainder theorem, consider $N_{L/K}(\alpha) \in \mathcal{O}_K \cap q_2 = p$; but

$$N_{L/K}(\alpha) = \prod_{\sigma \in \mathrm{Gal}(L/K)} \sigma(\alpha). \tag{$\dagger$}$$

On the other hand, by assumption, $\sigma(\alpha) \notin q_1$ for all $\sigma \in \mathrm{Gal}(L/K)$. Thus, $N_{L/K}(\alpha) \notin q_1$; but we already have $N_{L/K}(\alpha) \in p = q_1 \cap \mathcal{O}_K$, which is a contradiction. □

Note that the crucial application that L/K is Galois is given in ($\dagger$). The above theorem could fail "really badly" if L/K is not Galois.

Corollary 11.4. If L/K is Galois and q, q' are two primes lying over $p \subset \mathcal{O}_K$, then $e(q \mid p) = e(q' \mid p)$ and $f(q \mid p) = f(q' \mid p)$.

Proof. $e(q \mid p) = e(q' \mid p)$ follows from unique factorization. For f, since there exists $\sigma \in \mathrm{Gal}(L/K)$ such that $\sigma(q) = q'$, it induces an isomorphism $\mathcal{O}_L/q \cong \mathcal{O}_L/q'$. □

This means that all primes in a Galois extension have the same powers and indices of inertia.

Recall that, given L/K an extension of number fields, we say a prime ideal $p \subset \mathcal{O}_K$ is *ramified* if $e(q \mid e) > 1$, for some prime $q \subset \mathcal{O}_L$ lying over p. As an example, $\langle 2 \rangle$ is ramified in $\mathbb{Z}[i]/\mathbb{Z}$, since $(2) = (1+i)^2$.

Theorem. Let $p \in \mathbb{Z}$ be prime. Let K be a number field. p is ramified in $\mathcal{O}_K$ if and only if $p \mid \operatorname{disc}(\mathcal{O}_K)$.

We will prove this later.

Day 12: More on Ramification (Oct. 15, 2026)

We start by discussing Smith normal form (i..e, diagonalizing integer matrices). Let A be a $n \times n$ matrix over $\mathbb{Z}$; then A can be diagonalized by row and column operations.

Proof. Let $A = (a_{ij})$ and without loss of generality, $a_{ii} \neq 0$. For the first step, by row and column operations, we can arrange A so that $a_{11} \mid a_{1i}$ and $a_{11} \mid a_{i1}$ for all i. It is enough to show that a_{11} can be made to divide all other elements in the first row.

Suppose $a_{11} \nmid a_{12}$. Then $a_{12} = qa_{11} + r$, where $|r| < |a_{11}|$ (and nonzero r). In particular, $r = a_{12} - qa_{11}$. By column operations, we can make $|a_{12}| < |a_{11}|$ by permuting the first two columns. Now, by row and column operations, we can zero out all a_{i1} and a_{1i} for $i > 1$. This arranges A into a block diagonal matrix with a_{11} on the top left with a $(n-1) \times (n-1)$ block B in the bottom right, from which we may diagonalize B, and so on, inductively. □

By taking $d_i = \gcd(a_{ij})_{i,j=1}^{n-i+1}$, i.e., the greatest common divisor of all elements in the bottom right $(n-i) \times (n-i)$ block, we see that A can be diagonalized as

$$\begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix},$$

for which $d_1 \mid d_2 \mid \cdots \mid d_n$. We will leave it as an exercise to check that a version of this works for matrices over PIDs. Here, you will have to replace row and column operations by more general invertible matrices.

We return to ramification now.

Theorem 12.1. Let $p \in \mathbb{Z}$ be prime, and let $R = \mathcal{O}_K$ be a number ring. If p is ramified in R, then $p \mid \operatorname{disc}(R)$.

Proof. Let $q \subset R$ be a prime over p with $e(q \mid p) > 1$. This means $p = q^{e(q|p)}I = q(q^{e(q|p)-1}I)$, for which we will denote the bracketed portion J. J is divisible by all primes of R over p. Let $\sigma_1, \ldots, \sigma_n$ denote the embeddings of K in $\mathbb{C}$. Recall that there exists an extension $K \subset L$ such that $L/\mathbb{Q}$ is Galois; then $\sigma_1, \ldots, \sigma_n$ are automorphisms of L. Let $\alpha_1, \ldots, \alpha_n$ be an integral basis of R. We'll show that $p \mid \operatorname{disc}(\alpha_1, \ldots, \alpha_n)$.

Let $\alpha \in J \setminus pR$; then $J \subset r$ for every prime r over p. Pick $\alpha \in r$ for all primes r over p, so we have

$$\alpha = m_1\alpha_1 + \cdots + m_n\alpha_n.$$

Since $\alpha \notin pR$, it must be that $p \nmid m_i$ for some i. Without loss of generality, $p \nmid m_1$; we have

$$\operatorname{disc}(\alpha, \alpha_2, \ldots, \alpha_n) = m_1^2 \operatorname{disc}(\alpha_1, \ldots, \alpha_n)$$

by the change of basis theorem 6.1. Since $p \nmid m_1$, we see that

$$p \mid \operatorname{disc}(\alpha, \alpha_2, \ldots, \alpha_n) \iff p \mid \operatorname{disc}(\alpha_1, \ldots, \alpha_n).$$

α is contained in every prime of R over p, so α is contained in every prime if $S = \mathcal{O}_L$ over p. Fix a prime $r \subset S$ over p. We may observe that, for every field automorphism σ of L, we get that $\sigma(\alpha) \in r$. Equivalently, $\alpha \in \sigma^{-1}(r)$, which is a prime ideal over p; thus, $\sigma_i(\alpha) \in r$ for all i implies r contains $\operatorname{disc}(\alpha, \alpha_2, \ldots, \alpha_n)$. By the determinant formula, we see that

$$\operatorname{disc}(\alpha, \alpha_2, \ldots, \alpha_n) = \sum_{\tau \in S_n} \operatorname{sgn}(\tau) \prod_i \operatorname{Tr}(\beta_i \beta_{\tau(i)}),$$

where $\beta_1 = \alpha$ and $\beta_i = \alpha_i$ for $i > 1$. Note that the product above is divisible by $\mathrm{Tr}(\alpha\beta_{\tau(i)}) \in r$. Thus, $\mathrm{disc}(\alpha_1, \ldots, \alpha_n) \in r$, and is also an integer. Thus,

$$\mathrm{disc}(\alpha, \alpha_2, \ldots, \alpha_n) \in r \cap \mathbb{Z} = p\mathbb{Z},$$

so $p \mid \mathrm{disc}(\alpha, \ldots, \alpha_n)$. □

Corollary 12.2. Only finitely many primes of $\mathbb{Z}$ are ramified in a number ring R.

Proof. Every prime that ramifies has to be a divisor of a finite number. Clearly, finite numbers have finitely many primes in their prime factorization. □

Corollary 12.3. Let $\alpha \in \overline{\mathbb{Z}}$, and let $K = \mathbb{Q}(\alpha)$. Let f be any monic polynomial over $\mathbb{Z}$ such that $f(\alpha) = 0$. If p is a prime such that $p \nmid N^k f'(\alpha)$, then p is unramified in $\mathcal{O}_K$.

Proof. By the homework problem, for $d = \deg(\alpha)$,

$$\mathrm{disc}(1, \alpha, \alpha^2, \ldots, \alpha^{d-1}) \mid N^k f'(\alpha),$$

up to sign if f is minimal. The discriminant $\mathrm{disc}(\mathcal{O}_K) \mid \mathrm{disc}(1, \alpha, \ldots, \alpha^{d-1})$. □

Corollary 12.4. Let $K \subset L$ be an extension of number fields. Then only finitely many primes $q \subset \mathcal{O}_K$ are ramified.

Proof. Recall that if $q \subset \mathcal{O}_K$ is ramified in $\mathcal{O}_L$, then $q \cap \mathbb{Z} = p\mathbb{Z}$ is ramified. Finitely many primes $p \in \mathbb{Z}$ ramify in $\mathcal{O}_L$, and every prime $q \in \mathcal{O}_K$ lies over some prime $p \in \mathbb{Z}$. Moreover, for every prime $p \in \mathbb{Z}$, there are finitely many primes $q \subset \mathcal{O}_K$ over p. Thus, there can only be finitely many possibilities for ramified primes. □

What happens in quadratic number fields? Let $K = \mathbb{Q}(\sqrt{m})$, with m squarefree. How do primes split? Recall that, for $m = 2, 3 \pmod 4$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{m}]$. If $m = 1 \pmod 4$, then

$$\mathcal{O}_1 = \mathbb{Z}\left[\frac{1 + \sqrt{m}}{2}\right],$$

where $\mathrm{disc}(K) = m$. Let $p \in \mathbb{Z}$ be prime. What are the possibilities for prime factorization of $p\mathcal{O}_K$?

(i) If $p\mathcal{O}_K = q_1 q_2$, with q_1, q_2 distinct, and $f(q_1, p) = f(q_2, p) = 1$.

(ii) If $p\mathcal{O}_K = q^2$, and $f(q, p) = 1$.

(iii) If $p\mathcal{O}_K = q$, and $f(q, p) = 2$.

Theorem 12.5. Following the setup from above, where R is a number ring,

(i) If $p \mid m$, then $pR = (p, \sqrt{m})^2$.

(ii) If m is odd, then

$$2R = \begin{cases} (2, 1 + \sqrt{m})^2, & \text{if } m \equiv 3 \pmod 4, \\ \left(2, \frac{1+\sqrt{m}}{2}\right)\left(2, \frac{1-\sqrt{m}}{2}\right), & \text{if } m \equiv 1 \pmod 8, \\ \text{prime}, & \text{if } m \equiv 5 \pmod 8. \end{cases}$$

(iii) If p is odd and $p \nmid m$, then

$$pR = \begin{cases} (p, n + \sqrt{m})(p, n - \sqrt{m}), & \text{if } m = n^2 \pmod p, \\ \text{prime}, & \text{if } m \text{ is not a square mod } p. \end{cases}$$

Proof. (i) is an easy calculation. $(p, \sqrt{m})^2 = (p^2, p\sqrt{m}, m)$; $p \mid m$, so this lies in pR. m is squarefree and $p \mid m$ implies that $\gcd(p^2, m) = p$, where $p \in (p^2, m) \subset (p^2, p\sqrt{m}, m)$. Thus, $pR = (p, \sqrt{m})^2$.

The other cases where $p\mathcal{O}_K$ has 2 or p factors are proved the same way. It remains to show that (a) $2R$ is prime if $m \equiv 5 \pmod 8$ and (b) pR is prime if $p \nmid m$ and m is not a square modulo p. We prove the latter first. Let q be a prime over p. We need to show that $f(q \mid p) > 1$ (equivalently, it is equal to 2), i.e., we need to show that $R/q \notin \mathbb{F}_p$.

Observe that $x^2 - m$ has a root in R. implies that $x^2 - m$ has a root in R/q. By assumption, $x^2 = m$ doesn't have a root in $\mathbb{F}_p = \mathbb{Z}/p$; but it does have a root in R/q, so it means that R/q is not isomorphic to $\mathbb{F}_p$. The same idea follows through with (a).

$$f(x) = x^2 - x + \frac{1-m}{4}$$

has a root in R, but none in $\mathbb{F}_2$. Since $m \equiv 5 \pmod 8$, we see that $\frac{1-m}{4}$ is odd. f has a root in K because $\operatorname{disc}(f) = m$ is a square in K, so it has a root in $\mathcal{O}_K = R$, since f is monic with integer coefficients. □

We now give a preview of the next topic. Consider an extension of number fields $K \subset L$. Fix $\alpha \in L$ such that $L = K(\alpha)$, and fix a prime $q \subset \mathcal{O}_K$. Let us define some notation: for $h(x) \in R[x]$, denote $\overline{h}(x) \in (R/q)[x]$ be the reduction. Now, let $g(x) \in K[x]$ be a monic irreducible polynomial with $g(\alpha) = 0$. Now, let $g(x) \in \mathcal{O}_K[x]$, and consider $\overline{g}(x) \in (R/q)[x]$. R/q is a field (since prime ideals are maximal in Dedekind domains), so $(R/q)[x]$ is a PID. Thus, $\overline{g}$ factors as

$$\overline{g} = \overline{g}_1^{e_1} \dots \overline{g}_r^{e_r}.$$

The $\overline{g}_i$ are distinct irreducible monic polynomials. Next week, we will prove the following theorem.

Theorem. Suppose $p \nmid |\mathcal{O}_L/(R[\alpha])|$, where $p = q \cap \mathbb{Z}$. Note that $\mathcal{O}_L/(R[\alpha])$ is a quotient of abelian groups. Then $q\mathcal{O}_L$ factors as $r_1^{e_1} \dots r_m^{e_m}$, where r_i is the ideal

$$r_i = q\mathcal{O}_L + (g_i(\alpha)),$$

and $f(r_i \mid q) = \deg g_i$.

Day 13: Cyclotomic Fields (Oct. 22, 2026)

Let K, L be number fields. Let $n = |L : K|$, $R = \mathcal{O}_K$, and $S = \mathcal{O}_L$. Fix a nonzero prime $q \subset R$. Given $h \in \mathcal{O}_K[x]$, let $\overline{h}$ denote the corresponding element in $(\mathcal{O}_K/q)[x]$.

Fix $\alpha \in S$ (we want $\mathcal{O}_L \cong \mathcal{O}_K[\alpha]$), where α is a primitive element. Let $g \in K[x]$ be the monic irreducible polynomial for α over K; $\alpha \in \mathcal{O}_K$, so its conjugates also lie in $\mathcal{O}_K$, so $g \in \mathcal{O}_K[x]$.

We can consider that $\overline{g} \in (\mathcal{O}_K/q)[x]$ factors as $\overline{g} = \overline{g}_1^{e_1} \dots \overline{g}_r^{e_r}$, where $\overline{g}_i$ are distinct real monic polynomials.

Theorem 13.1. Suppose $p \nmid |\mathcal{O}_L : \mathcal{O}_K[\alpha]|$, where $p = q \cap \mathbb{Z}$. Then the prime decomposition of $q\mathcal{O}_L = \ell_1^{e_1} \dots \ell_r^{e_r}$, where $\ell_i = q\mathcal{O}_L + (g_i(\alpha))$. Also, $f(\ell_i/q) = \deg g_i$.

Proof. Let $f_i = \deg g_i = \deg \overline{g}_i$. We will show:

(i) For all i, $\ell_i = \mathcal{O}_L$, or $\mathcal{O}_L/\ell_i$ is finite dimensional of order $|\mathcal{O}_K/q|^{f_i}$.

(ii) $\ell_i + \ell_j = S$ for $i + j$.

(iii) $q\mathcal{O}_L \mid \ell_1^{e_1} \dots \ell_r^{e_r}$.

Assuming the above are all true, we can rearrange the ℓ_i such that $\ell_1, \dots, \ell_t \neq \mathcal{O}_L$, and $\ell_{t+1}, \dots, \ell_r = \mathcal{O}_L$. We then have that $\ell_1, \dots, \ell_t$ are prime ideals (per (i)), and contain q. (ii) implies that the ℓ_i are distinct, and (iii) implies that $q\mathcal{O}_L = \ell_1^{d_1} \dots \ell_r^{d_r}$, where the $d_i \leq e_i$ by unique factorization. We would then get $f(\ell_i/q) = f_i$ from (i).

In this manner,

$$n = f_1 d_1 + \dots + f_t d_t \leq f_1 e_1 + \dots + f_r e_r = (\deg \overline{g}_1)e_1 + \dots + (\deg \overline{g}_r)e_r = \deg g = n,$$

as α is a primitive element. We would then get that $t = r$, and $d_i = e_i$.

We now prove (i). We want to show that $\mathcal{O}_L/\ell_i$ is finite dimensional of order $|\mathcal{O}_K/q|^{f_i}$; we claim that

$$\frac{\mathcal{O}_L}{\ell_i} \cong \frac{(\mathcal{O}_K/q)[x]}{\overline{g}_i} = F_i.$$

Note that we have a surjective homomorphism $\varphi \colon \mathcal{O}_K[x] \to (\mathcal{O}_K/q)[x]/\overline{g}_i$; φ is surjective with kernel $\ker \varphi = q[x] + (g_i)$, so

$$\frac{\mathcal{O}_K[x]}{q[x] + g(i)} = \frac{\mathcal{O}_K[x]}{(q, g_i)} \cong F_i.$$

We also have that $\mathcal{O}_K[x] \to \mathcal{O}_L$, with $x \mapsto \alpha + \ell_i$, given by $\psi \colon \mathcal{O}_K[x] \to \mathcal{O}_L/\ell_i$. Now, $q \subset \ker \psi$ because ℓ_i lies over q, so $\ell_i \cap \mathcal{O}_K = q$, so $q \subset \ell_i$. Also, $g_i \in \ker \psi$ because $g_i(\alpha) = 0$. This means $(q, g_i) \subset \ker \psi$.

It follows that

$$\frac{\mathcal{O}_L}{\ell_i} \cong \frac{\mathcal{O}_K[x]}{(q, g_i)},$$

since (q, g_i) is a maximal ideal and $\ker \psi \neq \mathcal{O}_K[x]$ ($\psi(1) = 1$) when $ell_i \neq S$. Since ψ is surjective, we claim that $\mathcal{O}_L = \mathcal{O}_K[\alpha] + \ell_i$; then $p \in q \subset \ell_i$, and

$$p \nmid |\mathcal{O}_L/\mathcal{O}_K[\alpha]| \implies \mathcal{O}_L = \mathcal{O}_K[\alpha] + p\mathcal{O}_L.$$

This concludes (i).

We now prove (ii). Let $\overline{g}_i$ and $\overline{g}_j$ be distinct irreducible polynomials in $(\mathcal{O}_K/q)[x]$. By Bezout's, there exists $\overline{h}, \overline{k} \in (\mathcal{O}_K/q)[x]$ such that $\overline{g}_i\overline{h} + \overline{g}_j\overline{k} = 1$, i.e., $g_i h + g_j k = 1$ modulo $q[x]$. Evaluating at α, we have that $g_i(\alpha)h(\alpha) + g_j(\alpha)k(\alpha) = 1$ modulo $q\mathcal{O}_L$, so $1 \in (q, g_i(\alpha), g_j(\alpha)) = \ell_i + \ell_j$.

For (iii), let $\gamma_i = q_i(\alpha)$ and $\ell_i = (q, \gamma_i)$. We claim that $\ell_1^{e_1} \dots \ell_r^{e_r} \subset (q, \gamma_1^{e_1}, \dots, \gamma_r^{e_r})$; clearly, $q\mathcal{O}_L = (q) \subset (q, \gamma_1^{e_1}, \dots, \gamma_r^{e_r})$. Thus, it suffices to show that $\gamma_1^{e_1}, \dots, \gamma_r^{e_r} \in q\mathcal{O}_L$. Recalling that $\overline{q} = \overline{g}_1^{e_1} \dots \overline{g}_r^{e_r}$, we immediately get that $g = g_1^{e_1} \dots g_r^{e_r}$ modulo $q[x]$, which we may evaluate at α to get

$$\gamma_1^{e_1} \dots \gamma_r^{e_r} = 0 \pmod{q\mathcal{O}_L} \iff \gamma_1^{e_1}, \dots, \gamma_r^{e_r} \in q\mathcal{O}_L. \qquad \square$$

We now talk about cyclotomic fields. For $n \geq 1$, set $\zeta_n = e^{2\pi i/n} \in \mathbb{Z}^{\text{alg}}$. Then $\zeta_n^n - 1 = 0$. Let $K_n = \mathbb{Q}(\zeta_n)$. We denote the *primitive nth roots of* 1 to be ζ_n^b for $\gcd(b, n) = 1$. There are $\varphi(n)$ of these, where φ is Euler's totient function.

Note that every root of $x^n - 1$ is a power of ζ_n, so K_n is the splitting field for $x^n - 1$, and it is a finite extension, where $K_n/\mathbb{Q}$ is Galois. As an example, $K_1 = K_2 = \mathbb{Q}$, $K_3 = \mathbb{Q}(\sqrt{-3})$, and $K_4 = \mathbb{Q}(i)$.

Now, suppose $n = p^a$ is a prime power. Consider

$$\Phi_{p^a}(x) = \frac{x^{p^a} - 1}{x^{p^{a-1}} - 1} = 1 + x^{p^{a-1}} + \dots + x^{(p-1)p^{a-1}},$$

so ζ_{p^a} is a root of Φ_{p^a}, and $\deg \Phi_{p^a} = (p-1)p^{a-1} = \varphi(p^a)$.

As another eaxmple, $\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$, $\Phi_8(x) = x^4 + 1$, and $\Phi_9(x) = x^6 + x^3 + 1$.

Proposition 13.2. Φ_{p^a} is irreducible over $\mathbb{Q}$. $\left|K_{p^a} : \mathbb{Q}\right| = \varphi(p^a)$, and $\text{Gal}(K_{p^a}/\mathbb{Q}) = (\mathbb{Z}/p^a\mathbb{Z})^\times$.

Proof. We will prove this modulo p. First,

$$\Phi_{p^a}(x) = \frac{x^{p^a} - 1}{x^{p^{a-1}} - 1}.$$

In $(\mathbb{Z}/p\mathbb{Z})[x]$, it is equivalent to

$$\dots \equiv \frac{(x-1)^{p^a}}{(x-1)^{p^{a-1}}} \equiv (x-1)^{p^a - p^{a-1}} \equiv x^{p^a - p^{a-1}} - 1,$$

so $\Phi_{p^a}(x+1) = x^d$, where d is the degree of the above. By Eisenstein's criterion, we get that $\Phi_{p^a}(x+1)$ is irreducible ($\Phi_{p^a}(1) = p$). $\square$

Now, let $K = K_{p^a}$ and $\zeta = \zeta_{p^a}$. Let $R = \mathbb{Z}[\zeta]$, and write $\lambda = 1 - \zeta$, $d = \varphi(p^a) = |K : \mathbb{Q}|$.

Lemma 13.3. If $\gcd(b, p) = 1$, then $\frac{1-\zeta^b}{1-\zeta}$ is a unit in R.

Proof. Note that $1, \zeta, \zeta^2, \dots, \zeta^{d-1}$ forms a basis of $K/\mathbb{Q}$. We have

$$\left|\Delta(1, \zeta, \zeta^2, \dots, \zeta^{d-1})\right| = N(\Phi'_{p^a}(\zeta)) = N\left(p^a \frac{\zeta^{-1}}{\zeta^{p^{a-1}} - 1}\right).$$

Note that $N(p^a\zeta^{-1}) = p^{ad}$. Now,

$$N_{K/\mathbb{Q}}\left(\zeta^{p^{a-1}} - 1\right) = N_{K_p/\mathbb{Q}}\left(\zeta^{p^{a-1}} - 1\right) = p^{p^{a-1}},$$

as $\frac{|K:\mathbb{Q}|}{|K_p:\mathbb{Q}|} = p^{a-1}$. Thus,

$$\left|\Delta(1, \zeta, \zeta^2, \dots, \zeta^{d-1})\right| = p^{ad - p^{a-1}}. \qquad \triangle$$

Then there was some corollary about p being ramified and unramified in K.

Proof of 13.3, cont. $\mathcal{O}_K/\lambda \cong \mathbb{F}_p$. Indeed, $x \in \mathcal{O}_K$ yields $x = c_0 + \lambda x_1$, where $0 \leq c_0 < p$ and $x \in \mathcal{O}_K$ (by the claim). Repeating this d times yields

$$x = c_0 + c_1\lambda + \cdots + c_{d-1}\lambda^{d-1} + \lambda^d x_d,$$

where $0 \leq c_i < p$ for each c_i and $x_d \in \mathcal{O}_K$. We claim that $p\mathcal{O}_K = (\lambda)^d$, but this follows from the above. And then he gave up! □

Day 14: Primitive Roots of Unity (Nov. 3, 2026)

Last class, we started analyzing $K = K_{p^a}$, where p is prime. We have that

$$|K : \mathbb{Q}| = \varphi(p^a) = p^{a-1}(p-1),$$

and that $\mathrm{Gal}(K/\mathbb{Q}) = (\mathbb{Z}/p^a)^\times$. We also saw that

$$\Phi_{p^a}(x) = \frac{x^{p^a} - 1}{x^{p^{a-1}} - 1}$$

is irreducible. Now, define $\lambda = 1 - \zeta_{p^a}$; since

$$N(\lambda) = \prod_{b \in (\mathbb{Z}/p)^\times} (1 - \zeta^b) = \Phi_{p^a}(1) = p, \tag{$*$}$$

it follows that (λ) is a prime ideal. Note that

$$\left(\prod_{b \in (\mathbb{Z}/p)^\times} \frac{1-\zeta^b}{1-\zeta} \right) (1-\zeta)^{\varphi(p^a)} = p$$

by $(*)$, and so $\frac{1-\zeta^b}{1-\zeta} = 1 + \zeta + \cdots + \zeta^{b-1} \in \mathcal{O}_K$ is a unit. We now want to show that $\frac{1-\zeta}{1-\zeta^b} \in \mathcal{O}_K$. First, $p \nmid b$, so there exists c such that $bc = 1$ modulo p^a, and so $\zeta = (\zeta^b)^c$. We also have that

$$\frac{1-\zeta}{1-\zeta^b} = \frac{1-(\zeta^b)^c}{1-\zeta^b} = 1 + \zeta^b + \cdots + (\zeta^b)^{c-1} \in \mathcal{O}_K,$$

whence $p = u\lambda^{\varphi(p^a)}$, where $u \in \mathcal{O}_K^\times$. Writing $d = \varphi(p^a)$, we have that $p\mathcal{O}_K = \lambda^d \mathcal{O}_K$, where (λ) was seen to be prime as before, so $e((\lambda) \mid p) = d$ and $f((\lambda) \mid p) = 1$. Thus, $\mathcal{O}_K/(\lambda) \cong \mathbb{F}_p$. Last time, we saw that

$$\left|\Delta(1, \zeta, \ldots, \zeta^{d-1})\right| = p^{p^{a-1}(a(p-1)-1)} = p^{a\varphi(a^p) - \frac{\varphi(a^p)}{p-1}} = p^{ad - \frac{d}{p-1}}.$$

In particular, all other primes are unramified in K.

Theorem 14.1. $\mathcal{O}_K = \mathbb{Z}[\zeta]$.

Proof. Since $\mathcal{O}_K/(\lambda) \cong \mathbb{F}_p$, for $x \in \mathcal{O}_K$, we have that $x = c_0 + \lambda x_1$, with $x_1 \in \mathcal{O}_K$ and $c_0 \in \mathbb{Z}$ with $0 \le c_0 < p$. Repeating this d times, we get

$$x = c_0 + c_1\lambda + \cdots + c_{d-1}\lambda^{d-1} + \lambda^d x_d,$$

where $x_d \in \mathcal{O}_K$, and the first d terms lie in $R \coloneqq \mathbb{Z}[\zeta]$. Since $(\lambda^d) = (p)$, it follows that $\lambda^d x_d = px'_d$, where $x'_d \in \mathcal{O}_K$.

The upshot is that $\mathcal{O}_K = R + p\mathcal{O}_K$. Now, let $T = \mathcal{O}_K/R$ be an abelian group. T is finite and its order is a power of p. Also, note that $p\colon T \to T$ is a bijection (where p denotes multiplication by p), so for $\overline{x} \in T$, where $x = x_0 + px_1$ with $x_0 \in R$, we have that $\overline{x} = p\overline{x}_1$. We get a contradiction out of this; if $|T|$ is a power of p, then $p\colon T \to T$ is not injective, so $|T| = 1$, i.e., $\mathcal{O}_K = R$. □

We now discuss the general roots of 1.

Theorem 14.2. For all u, we have that $|K_n : \mathbb{Q}| = \varphi(n) =: d$, where $\mathcal{O}_{K_n} = \mathbb{Z}[\zeta_n]$, and

$$\left|\Delta_{K_n}\right| = \frac{n^d}{\prod_{p|n} p^{\frac{d}{p-1}}}.$$

If $p \mid n$, then the power of p in $\left|\Delta_{K_n}\right|$ is $ad - \frac{d}{p-1}$, where a is the largest choice such that $p^a \mid n$.

Proof. We will induct on the number of distinct primes dividing n. For the base case of 1 prime, we have $n = p^a$, which we've already resolved. Now, write $n = p^a m$, where $\gcd(p, m) = 1$ with $m > 1$. Let $E = K_{p^a}$ and $F = K_m$. Observe that $EF = K_n$; K_n is generated by ζ_n, which is a primitive nth root of unity. Since p and m are coprime, we have that $um + vp^a = 1$ for some $u, v \in \mathbb{Z}$, so $\zeta_{p^a}^u \zeta_m^v$ is a primitive nth root of unity as well, so $K_n \subset EF$.

Conversely, ζ_n^m is a primitive p^ath root of unity, and $\zeta_u^{p^a}$ is a primitive mth root of unity. This means $E \subset K_n$ and $F \subset K_n$. △

Before we finish the proof, let us make some observations first (which are left as an exercise).

(i) $E \cap F = \mathbb{Q}$.

(ii) $|EF : \mathbb{Q}| = |E : \mathbb{Q}|\,|F : \mathbb{Q}| = \varphi(p^a)\varphi(m) = \varphi(n)$.

Proof of 14.2, cont. The upshot being that $|K_n : \mathbb{Q}| = \varphi(n)$. By induction, Δ_{K_m} and $\Delta_{K_{p^a}}$ are coprime. This yields

(i) $\mathcal{O}_{K_n} = \mathcal{O}_{K_m}\mathcal{O}_{K_{p^a}} = \mathbb{Z}[\zeta_m, \zeta_{p^a}] = \mathbb{Z}[\zeta_n]$,

(ii) $\Delta_{K_n} = \Delta_{K_m}^{|K_{p^a}:\mathbb{Q}|} \cdot \Delta_{K_{p^a}}^{|K_m:\mathbb{Q}|}$.

The power of p in Δ_{K_n} is given by

$$\Delta_{K_{p^a}} = a p^{\varphi(p^a) - \frac{\varphi(p)}{p-1}} = \left(a\varphi(p^a) - \frac{\varphi(p^a)}{p-1}\right)\varphi(m) = a\varphi(n) - \frac{\varphi(n)}{p-1} = ad - \frac{d}{p-1}. \qquad \square$$

Corollary 14.3. For all n, the polynomial

$$\Phi_n(x) = \prod_{\substack{1 \le b \le n \\ \gcd(b,n)=1}} (x - \zeta_n^b)$$

is the minimal polynomial of ζ_n. In particular, $\Phi_n(x) \in \mathbb{Z}[x]$ and is irreducible over $\mathbb{Q}$. Moreover, $\mathrm{Gal}(K_n/\mathbb{Q}) = (\mathbb{Z}/n)^\times = \sigma_b(\zeta_n) = \zeta_n^b$, and

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$